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Conventional Train Control-VF.CONCEPT

SFRS for Zefiro-Europe

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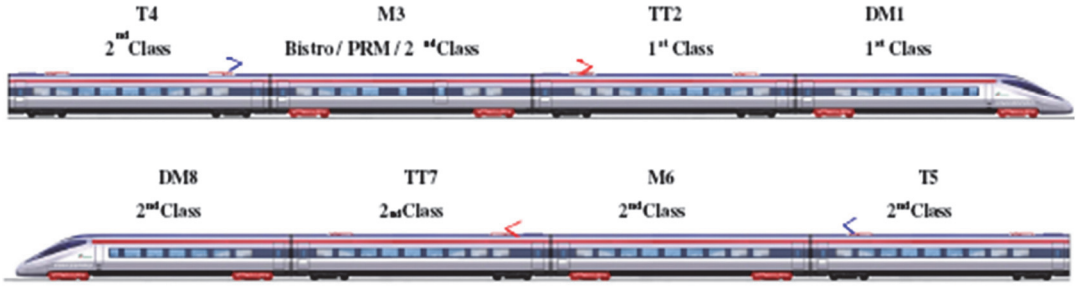
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1 Introduction

This document shall provide a description of the functionality for the V300 Zefiro Train Control System including ordinary train control (OTC) by conventional relay logic and train lines, TCMS electronics and Sub-system control unit.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

ATP	Automatic Train Protection
AX	Analogue Input/output unit
BCU	Brake control unit
CCU-D	Central computing unit diagnositcs
CCU-O	Central computing unit operational
CD	Circuit Diagram
CDR	Conceptual design review
DBV	Discharge Brake Valve
DCU/M/L/F/A	Drive control unit motor/line/frequency/auxiliary converter
DIS	Drivers Indicated Speed
DNRA	Detection of non rotating axle
DSD	Drivers safety device
ETB	Ethernet Train Backbone
EVENT	Diagnostics event that will be stored in the diagnostics database on the CCU-D
GW	Gateway (WTB/MVB)
HVAC	Heating Ventilation and air condition
HV	High voltage
HW	Hardware
HW-HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Intelligent Display Unit
I/O	Input/output
LCB	Line Circuit Breaker
LED	Light Emitting Diode
MCM	Motor converter module
MIO	Modular Digital input/Output unit
MVB	Multifunctional vehicle bus
NVRAM	Non Volatile Random Access Memory
OTC	Ordinary train control(relay logic/hardware by circuit diagram)
PCU	Propulsion Control Unit

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SFRS	System Functions Requirement Specification
SIL	Safety Integrity Level
TBD	To be defined
TCMS	Train Control and Management System
TL	Train lines
TSI	Train Specification for Interoperability
UIC	Union Internationale des Chemin (International Union of Railways)
WSP	Wheel slide protection
WTB	Wire Train Bus

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2 System Overview

2.1 Train Control Principles

The train will have a redundant two bus battery system that will affect all other electrical system/functionalities assembled.
The Train Control System is a combination of

1. **Ordinary Train Control** (OTC) comprising relay logic and train control lines
2. **TCMS electronics**, comprising hardware, software and communication networks
3. **Sub-system control unit**

The safety integrity level (SIL) is the major design constraint on where and how a function is realised, in the Train Control System, using a safe computer in the TCMS electronics and/or by the OTC relay logic.
All safety critical and safety related functions will be controlled by OTC or a safe computer.
The identified vital functions are realised using redundant train line pairs.
Train lines that if faulty will stop operation are possible to by-pass.
All vital functions related to TSI requirements shall be duplicated or protected to cope with a fire incident in a way that the train is always able to perform an evacuation drive.
The TCMS electronics will be used for diagnostics. It will also be used to control functions that are not safety related.

2.2 System Configuration

Train supervision and indications:

All safety critical and safety related functions will be controlled or supervised via OTC or a safe computer.
The identified vital functions are realised using redundant or fire safe train lines.
Train lines that if faulty will stop operation are possible to by-pass.

All vital functions will be duplicated to cope with a fire incident in a way that the train is always able to drive with limited performance (emergency operation).

The TCMS will be used for diagnostics and monitoring. It will also be used to control functions that are not safety related.

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3 Functional requirements

3.1 Train Control

3.1.1 Cab Activation/Deactivation

The cab is activated by the driver or the crew personal moving the key switch on the Active position. This function works only if the train has been activated. Note: The train is activated
after the pushbutton in the electrical cabinet in DM-car has been pushed and the Train Activation procedure did not fail.

A single Train Line is provided to inform that one cab is active (in single or multiple configuration).

3.1.1.1 Design

Cab Activation
Interlock must be implemented in order to prevent two cabs from being activated at the same time. If another cab activation attempt is done while a cab is activated, an advisory message is generated by TCMS.
In case two driver's key switch are rotated in active position when the train is deactivated, the TCMS cannot monitor the inconsistency since the 110 Vdc is not available. If in this condition the train active procedure is performed, no cabs must be activated and a fault need to be sent to the IDU and the "TCMS busy" lamp needs to be illuminated steadily.

The TCMS supervises the status of the Cab Active Train line and of each desk key.
In case of any failure or inconsistency (eg two cabs active at the same time) traction block will be set until the train is back in normal condition or the block is reset using the IDU in maintenance mode.

Cab Deactivation
The cab shall not be deactivated when speed >0km/h (note train is not standstill). If the key is turned when the train is running the Emergency Brake shall be applied and the line voltage shall be deactivated (LCB open and panto down).

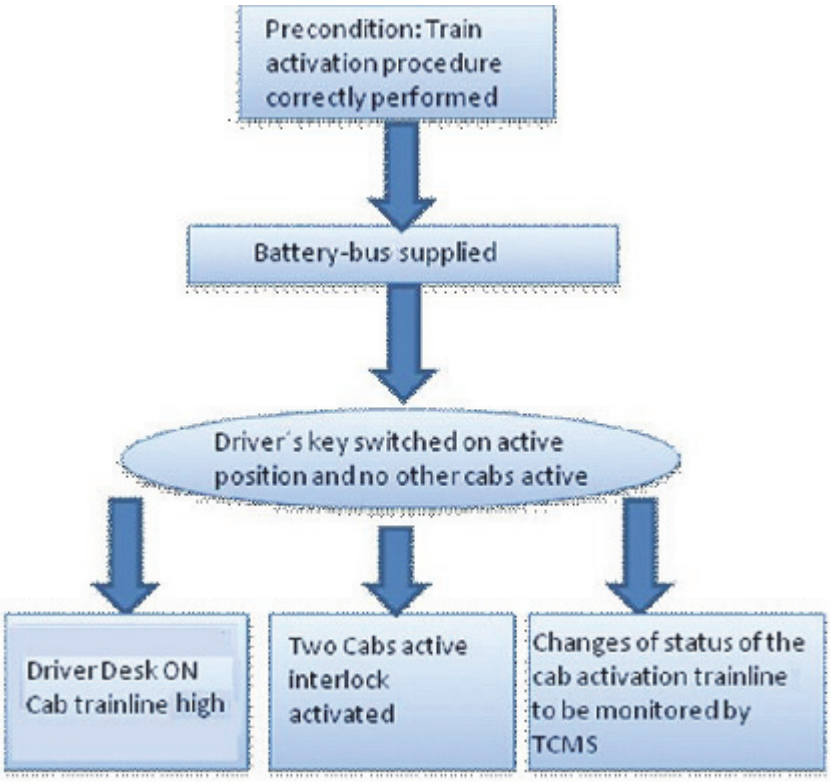
When the driver de-activates the cab then the proper shut down sequence for line voltage will be performed unless Parking mode is activated (see Parking mode section).

The status of the cab activation trainline is supervised by TCMS.

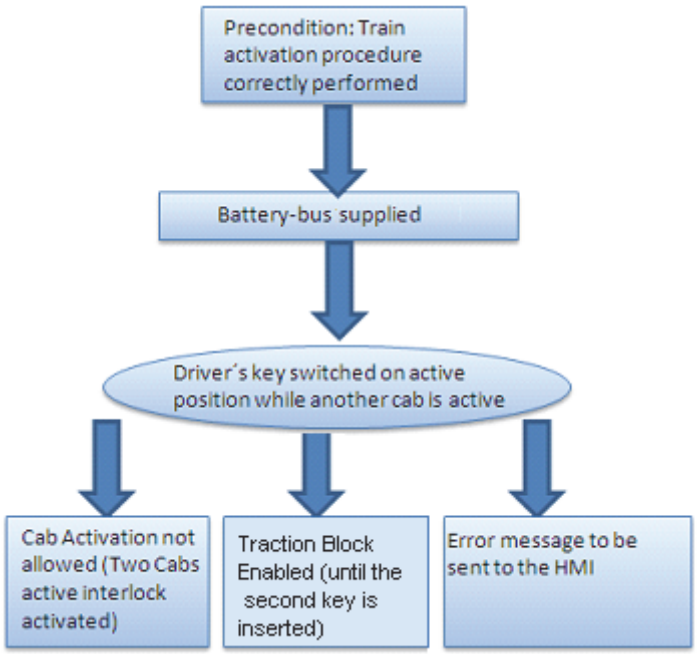
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Cab Activation



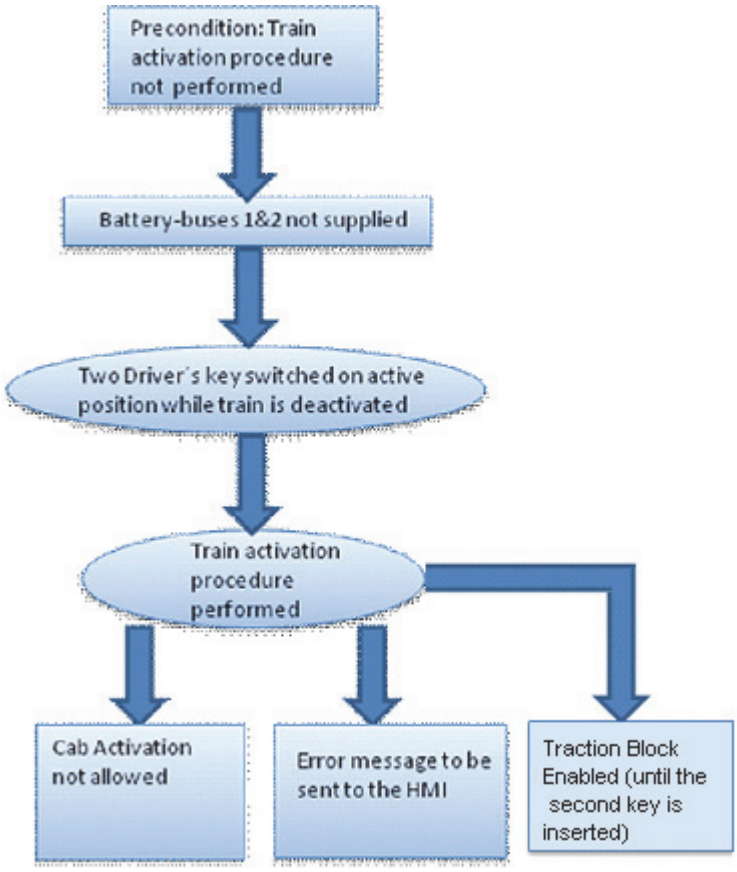
Cab Activation not allowed a)



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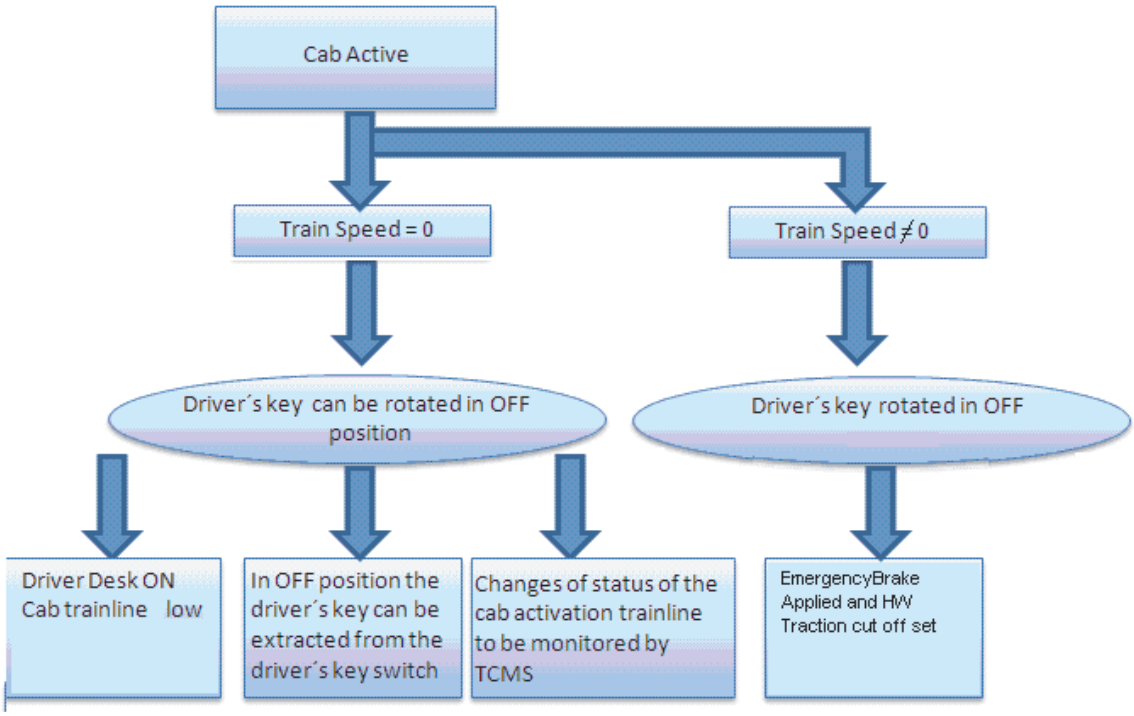
Cab Activation not allowed b)



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Cab Deactivation



3.1.1.1.1 Normal condition

1) Cab Activation: In normal condition driver rotate the driver´s key on the active position and the cab trainline shall be high.
2) Cab Deactivation: In normal condition driver rotate the driver´s key on the off position and the cab trainline shall be low.

3.1.1.1.2 Failure cases

1) The cab trainline is not high after the driver´s key has been turned on active position. Train will be immobilised.
2) The cab trainline is not low after the driver´s key has been turned on OFF position. Driving the train from opposite cab will not be possible.
3) Cab is activated when no driver´s key is in the active position.
4) Fault of one cab activation train line trainline.

3.1.1.1.3 Failure detection

1) Performed by the driver (driver desk not active)
2) Performed by the driver (driver desk still active)
3) In both both cabs are active a fault message will be displayed on the IDU
4) Fault message on the IDU

3.1.1.1.4 Failure handling

1) Retire or don´t put the vehicle in service. Immediate corrective maintenance required.
2) Retire or don´t put the vehicle in service. Immediate corrective maintenance required.
3) Maintenance needed before putting the vehicle in service again.
4) Maintenance needed at the end of the service.

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3.1.1.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.347
Description	Two Position: 0 = Cab off (stable position) 1 = Cab activated (stable position). It is possible to extract the key only in OFF position
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1007	Cab shall be activated/de-activate by means of two signals "Cab Activated" and "Cab Off". The signals are managed by the driver by means of dedicated key switch (2 stable position switch).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_09.03.Zefiro-Europe.CONCEPT.1109	Cab is activated/de-activate by means of two signals "Cab Activated" and "Cab Off". The signals are managed by the driver by means of dedicated key switch (2 stable position switch).	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.953	If the cab is attempted to be de-activated by the driver's key when train speed > 0km/h the Emergency Brake Loop shall be opened (EB applied) and the Traction Safe line shall be de-energized	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1279	If the cab is attempted to be de-activated by the driver's key when train speed > 0km/h an advisory event shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.174
Description	Cab activation is made by means of the driver’s key.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1006	Cab is activated when the signal "Cab Activated" is high (this is done by the driver moving the key on the Cab activated stable position).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_09.03.Zefiro-Europe.CONCEPT.1342	A cab shall be considered as activated when the 'Cab activated' input is set OR all the following conditions apply for at least 3 seconds: – The "Any cab active' input is set – No "Cab activated" input is set anywhere in the train – The drivers key is in active position	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1596	In case a second cab is attempted to be enabled or disabled, TCMS shall consider first activated cab (key inserted and rotated) as really enabled. This is valid also in case of multiple configuration.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.175
Description	Activation of a second cab in a train shall be prevented.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.948	Interlock shall be implemented in order to prevent two cabs from being activated at the same time.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1436	If another cab is attempted to be activated (desk key turned) then Traction Block shall be set by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.950	Any cab activation attempt in another non active cab shall be indicated in the cab where the activation is attempted.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

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2F_09.03.Zefiro-Europe.CONCEPT.951	The status of the 'driver's key' input, 'cab active' input (LCA and LCA-R relays) and the 'any cab activated trainline' input shall be supervised by TCMS for plausibility. If an inconsistency is detected an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1253	The 'Cab active' position of the driver's key shall be indicated to TCMS. The status of Cab Active relays (LCA) shall be provided to TCMS reading the series of NC and NO contacts separately. The status of the "any cab actived" trainlines shall be indicated to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.222
Description	Sufficient brake shall be applied e.g. by opening the emergency brake loop in case the cab is deactivated. This shall avoid roll back in case of not occupied cab.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.206	When no cab is active in the train the emergency brake loops shall be opened (Emergency Brake Applied).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1254	If there is no active cab in the train and the Emergency brake is not ordered (EB loop is energised) then an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1418	If there is no active cab in the train then Traction block shall be applied.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2 Demand of Driving Direction

3.1.2.1 Design

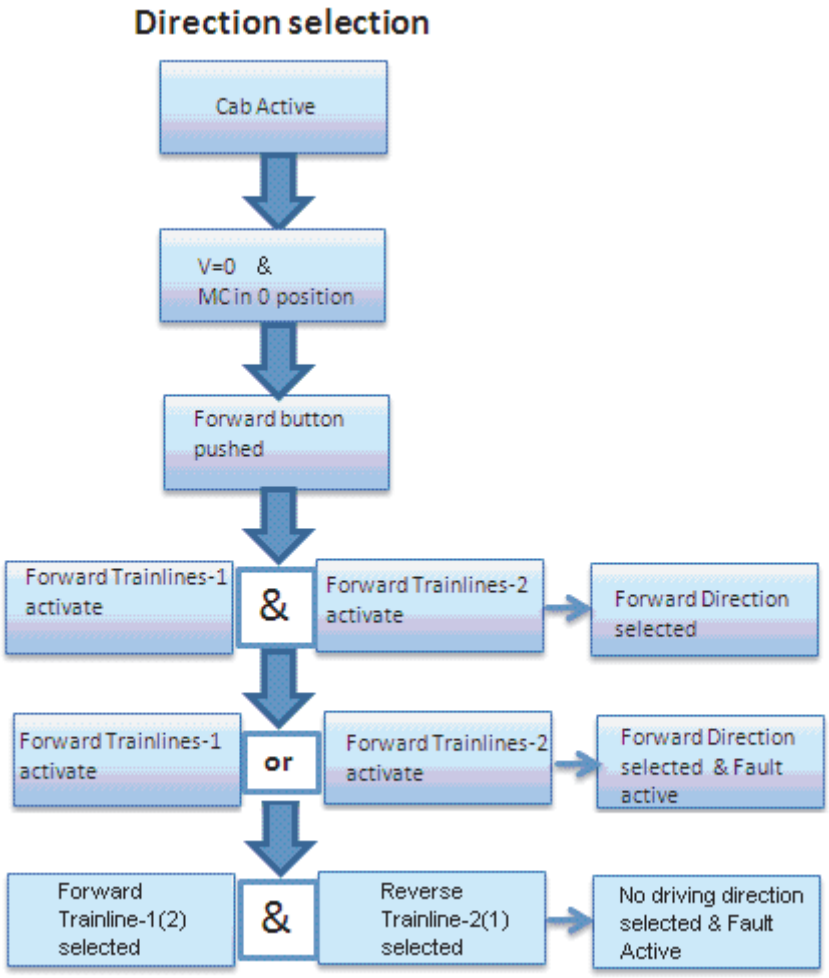
The driving direction signal is in normal operation transferred by trainlines. At failure a redundant parallel trainline is used.

It will be not possible to pass directly from Forward to Reverse direction. The driver must go from forward to neutral by pushing the neutral push-button at train standstill and with the master controller in position 0. Then from Neutral the **Reverse** direction can be selected by pushing the reverse push-button, similar behaviour need to be foreseen when it is needed to go from Reverse to Forward.

The TCMS will be used for monitoring. The forward/reverse trainlines will be wired into the DCUM for motor converter modules to control the driving direction.

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Same is valid for reverse direction selection.

When driver select Reverse Direction speed limitation is applied.

3.1.2.1.1 Normal condition

1) After pushing the Forward pushbutton the required direction is selected
2) After pushing the Reverse pushbutton the required direction is selected

3.1.2.1.2 Failure cases

1) After pushing the Forward pushbutton the required direction is not selected
2) After pushing the Reverse pushbutton the required direction is not selected
3) After pushing the Forward(Reverse) pushbutton the opposite direction is selected

3.1.2.1.3 Failure detection

1)
   - Performed by the driver (Forward pushbutton is not lit after selection)

2)
   - Performed by the driver (Reverse pushbutton is not lit after selection)

3)
   - Performed by the driver

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3.1.2.1.4 Failure handling

1) Retire or don´t put the vehicle in service
2) Retire or don´t put the vehicle in service
3) Retire or don´t put the vehicle in service

3.1.2.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.230
Description	Driving in reverse direction by system failure shall be avoided.
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.965	It shall not be possible to change direction directly from forward to reverse or from reverse to forward. Neutral must be select before selecting direction by pushing the neutral push-button at train standstill and when the master controller is not in traction position. The direction can then be selected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_09.03.Zefiro-Europe.CONCEPT.437	If change of driving direction is attempted when the train is not standstill or when master controller is in traction position, an advisory event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.850	The driving direction shall be sent to propulsion control via trainlines in the whole train according to the schema below: Car / Cab active in DM1 and Forward direction selected / Cab active in DM8 and Forward direction selected DM1 / Forward / Backward M3 / Forward / Backward M5 / Backward / Forward DM8 / Backward / Forward Car / Cab active in DM1 and Reverse direction selected / Cab active in DM8 and Reverse direction selected DM1 / Backward / Forward M3 / Backward / Forward M5 / Forward / Backward DM8 / Forward / Backward If Neutral is selected on the driver's desk neither forward or backward shall be sent to the Propulsion control.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_09.03.Zefiro-Europe.CONCEPT.848	Propulsion control shall supervise the direction signals received by trainlines and take actions according to below table: Forward Trainline 1 / Forward Trainline 2 / Reverse Trainline 1 / Reverse Trainline 2 / Forward / Reverse / No driving direction / Fault indication 0 0 0 0 x 0 0 0 1 x x 0 0 1 0 x x 0 0 1 1 x 0 1 0 0 x x 0 1 0 1 x x 0 1 1 0 x x 0 1 1 1 x x 1 0 0 0 x x 1 0 0 1 x x 1 0 1 0 x x 1 0 1 1 x x 1 1 0 0 x 1 1 0 1 x x 1 1 1 0 x x 1 1 1 1 x x	Derived Requirement	4S_05.-.Propulsion		N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1112	An event shall be displayed on the IDU if reverse direction has been selected.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.148
Description	Change of direction shall only be allowed when: – at standstill (v=0) – master controller in position "0"
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.515	The "forward direction" pushbutton, "reverse direction" pushbutton, "neutral" pushbutton shall be connected to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1440	After the cab activation it shall not be possible to select any direction if the Master Controller is in Traction position. If a direction is selected while the Master controller is in Traction the TCMS shall apply Traction Block until the handle is moved in "0" or "braking" position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1280	The status of below input shall be sent to TCMS from each PCU in the train: – Forward trainline input 1 – Forward trainline input 2 – Reverse trainline input 1 – Reverse trainline input 2	Derived Requirement	4S_05.-.Propulsion		N/A

ID	2F_09.03.Zefiro-Europe.TRS.149
Description	Maximum reverse speed shall be limited to 40km/h.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.69	If driving in reverse direction has been demanded, the train speed shall be limited to 40km/h, the value shall be adjustable as parameter stored in the CCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.61
Description	When a cab will be activated this cab will be defined as head of the train. From this results the normal driving direction (forward direction). For shunting trips it is be possible to select reverse direction.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1009	The head of the train is defined according to the cab activated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.62
Description	The active cab shall always define the forward direction. The driving drection shall be actively chosen by the driver using a "forward" or a "Reverse" button in the driver's desk.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.972	Forward direction of the train shall always be determined according to the active cab.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

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ID	2F_09.03.Zefiro-Europe.TRS.351
Description	Direction selector pushbuttons Forward:Lit when selected, Neutral: Lit when selected (neutral selected by default when at the desk activation) Backward: Lit when selected
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.969	Three buttons are available on the direction selection panel: Forward: Lit when selected Neutral: Lit when selected (neutral selected by default when at the desk activation) Backward: Lit when selected	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.971	When the cab is activated "neutral" shall be set.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1344	When the cab is activated an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1398	When the cab is activated and direction is not "neutral" a Fault message shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1415	If no valid direction is set Traction block shall be applied.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1416	If the direction is changed at speed >0 km/h then Traction block shall be applied. The Traction block shall be reset when master control handle is not in traction position AND (the evaluated direction is equal to real direction OR train speed = 0).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.3 Traction/Brake reference

3.1.3.1 Design

3.1.3.1.1 Master Controller and Speed Lever

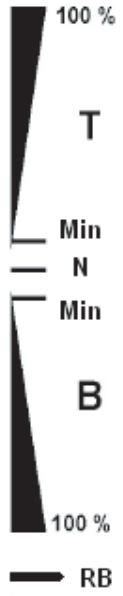
Each cab is equipped with a Master Controller and a Speed Lever controller.

Master Controller: Two redundant encoders with resolution of 6bit each one are used in the Master controller to determine the demanded traction/brake reference.
The encoders are connected directly to TCMS

A set of microswitch is included in the controller to provide interfaces with:
- Traction/Not Brake trainlines (see Emergency Operation)
- Emergency Brake Loop

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(*)	#Pos	Position [%]	Description
	6	T 100 %	*Maximum Traction Effort*
	5	T Min %	*Minimum Traction Effort*
	4	N	***Traction** can only be applied by **pressing the knob** on top of handle*
	3	B Min %	*Minimun Braking Effort*
	2	B 100 %	*Maximum Braking Effort*
	1	EB	*Emergency Braking Position*

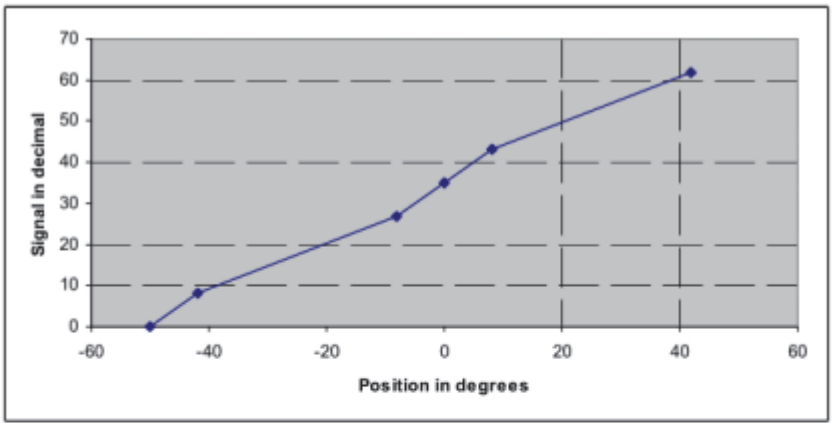
Switching table for master controller micro-switches

Micro-Switch ? / Position ?	S11	S12	S13	S14	S15- nc	S15- no	S16- nc	S16- no
T MAX	1	1	1	1	1	0	1	0
T MIN	1	1	1	1	1	0	1	0
N	0	0	1	1	1	0	1	0
B MIN	0	0	0	0	1	0	1	0
B MAX	0	0	0	0	1	0	1	0
EB	0	0	0	0	0	1	0	1

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The encoder output is grey coded as shown in the table below



LEVER POSITION	POSITION IN DEGREES	OUTPUT SIGNAL IN DECIMAL	OUTPUT SIGNAL GRAY CODE					
			D6	D5	D4	D3	D2	D1
Rapid Brake	-50°	0	0	0	0	0	0	0
Max Braking	-42°	8	0	0	1	1	0	0
Min Braking	-8°	27	0	1	0	1	1	0
Coasting	0°	35	1	1	0	0	1	0
Min Traction	8°	43	1	1	1	1	1	0
Max Traction	42°	62	1	0	0	0	0	1

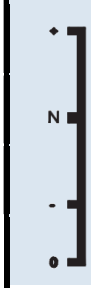
The Master controller also provides a signal to the vigilant device pushing the top of the handle.
Speed Lever: The speed lever can be used by the driver to select a speed reference. The train will then be limited to this speed by TCMS.
The speed lever positions are connected directly to TCMS.

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(*)	#	Position	Description
	4	(+)	*Spring-loaded return to Position 3*
	3	N	*Predefined position:* *To **increase** target speed push forward the lever and press the knob on the top of the handle* *To **decrease** target speed, **no specific action** required*
	2	(-)	*Spring-loaded return to Position 3.*
	1	0	*Predefined position, semi-locked. ASC off*

Switching table for Speed controller

Micro-Switch ? / Position ?		S101	S102	S103	S104	S105	S106
	Speed +	0	0	0	0	1	1
	N	0	0	1	1	1	1
	Speed -	0	0	1	1	0	0
	0	1	1	1	1	1	1

3.1.3.1.2 Normal operation

In normal operation either the Master controller or the Speed Lever can be used to adjust the amount of requested traction/braking demand relevant to available traction braking effort.

The driver can change the mode to a speed set reference to drive the train to a specific speed. The Master controller will also be used to request the maximum amount of power that will be used to achieve the set speed reference.

During normal operation the driver can change from traction to speed mode by means of the dedicated speed lever acknowledgment button which shall be pressed while the master controller is in coasting position ("N") and the speed lever in "N" position (other combination will not allow the change from traction mode to speed mode).
When the speed mode has been activated, the actual speed of the train shall be used as reference speed. The driver can adjust the reference speed in step of 1 km/h or in step of 5 km/h. It depends on the time the driver keeps the speed lever in + or – position.

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3.1.3.1.3 Failure cases (master controller)

a) Traction is demanded when not requested
b) Braking is demanded when not requested
c) Traction is not demanded when requested
d) Braking is not demanded when requested
e) One encoder is not working

3.1.3.1.4 Failure detection (master controller)

a) By the driver
b) By the driver
c) By the driver
d) By the driver
e) Message on the IDU

3.1.3.1.5 Failure handling (master controller)

a) Operate the 'Master controller back up mode' switch (sealed) and use the speed lever to demand a traction reference. Switch Brake mode to "Indirect brake" and use DBV to set brake reference. Maintenance needed at the end of the service.
b) Operate the 'Master controller back up mode' switch (sealed) and use the speed lever to demand a traction reference. Switch Brake mode to "Indirect brake" and use DBV to set brake reference. Maintenance needed at the end of the service.
c) Operate the 'Master controller back up mode' switch (sealed) and use the speed lever to demand a traction reference. Switch Brake mode to "Indirect brake" and use DBV to set brake reference. Maintenance needed at the end of the service.
d) Operate the 'Master controller back up mode' switch (sealed) and use the speed lever to demand a traction reference. Switch Brake mode to "Indirect brake" and use DBV to set brake reference. Maintenance needed at the end of the service.
e) Complete the service. Maintenance needed at the end of the service.

3.1.3.1.6 Failure cases (Speed lever)

a) Speed of train is greater than the speed set reference
b) Speed of train is lower than the speed set reference

3.1.3.1.7 Failure detection (Speed lever)

a) By driver or ATP demands brake if the train speed is over the allowed track speed
b) By the driver

3.1.3.1.8 Failure handling (Speed lever)

a) The train will try to regulate the speed by braking to the speed set reference. The driver can also select a brake position of the Master controller. If the speed set reference is the same or close to that of the ATP speed limit the ATP will brake the train. Maintenance required.
b) The driver can increase the power requested by the Master controller or remove the speed set reference mode and drive only with the Master controller.

3.1.3.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.197
Description	Two controllers, the ordinary master controller and the speed controller work independently.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1010	Master controller and Speed Set Lever are two independent units.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1196	Master controller and Speed Set Lever are two independent units.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

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ID	2F_09.03.Zefiro-Europe.TRS.336
Description	The traction demand level shall be distributed from the active cab to all DCUM along the train.
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1149	The positive effort shall be equally shared among the available traction inverter while the negative effort shall be distribute according to the Blending law (Blending law description is not part of this SFRS).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.794	The traction demand shall be forwarded and distributed to the PCU in the motor cars.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.795	If traction demand is > 0% it shall be indicated to the PCU that train is in traction.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.796	If traction demand is < 0% it shall be indicated to the PCU that train is in braking.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.797	If friction brake is applied in the train then the PCU shall be notified.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

In case of single hardware failure in TCMS the traction shall not be reduced more than 12,5% which correspond to the traction capability of two MCMs.

ID	2F_09.03.Zefiro-Europe.TRS.338
Description	No traction force shall be applied if not demanded.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.792	If the Master controller is not in a traction position or the Emergency traction trainline is not active or traction is not demanded by the Speed set lever in Master controller back-up mode then the traction demand to all PCU's in the train shall be set to 0%.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.359
Description	The master controllers brake position has priority even if the speed controller is not in “0” position. This because braking has always priority.
Safety level	N/A

The train can be driven in Power reference mode or Speed set reference mode.
Power reference mode is when only the power reference from the master control controller demands a specific amount of traction or brake percentage based on available traction or braking effort.
Speed set reference mode is when the speed is selected by the speed set lever. The power for achieving the speed set reference is then demanded by the power reference position of the Master controller.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1155	It shall not be possible to select a power or speed set reference mode if traction block is active and the master controller is not in the coast position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1156	In Power reference mode operation both requested and applied torques shall be displayed on IDU. In Speed reference mode the requested speed (target speed) shall be displayed on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1157	The control of the traction/brake demand shall be automatically set to Power reference mode at cab activation.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1158	To select speed set reference mode the speed lever shall be moved to "N" position while the master controller is in the 'N' position (coasting).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1159	When 'speed reference mode' is selected the current train speed shall be the used as the target one.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1532	After the speed control mode has been enabled, the traction position of the master controller shall limit the power reference to achieve the required speed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1160	To return to power reference mode from speed set reference mode, the speed lever shall be moved in "0" position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1161	Switching between modes can be done at any time, regardless of speed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1408	When speed reference mode is reset Traction block shall be applied until master controller is in coast or brake position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1162	The power or speed set reference shall be shown as a flashing “0” on the IDU when there is a traction block but the requested torque is not negative.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1163	If a neutral section or separation section is detected the power brake reference shall be set to 0 or a negative value.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1164	When neutral section or separation section is reset the power brake reference shall be resumed (traction is allowed).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1165	The IDU shall show the selected speed set reference or power reference during the whole neutral section or separation section passage even though the power brake reference is temporary set to 0 or a negative value.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1166	The master controllers brake position has priority even if the speed controller is not in “0” position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.187
Description	A speed controller for speed set (cruise control) shall be provided.Positions 4: increase speed set point 3: neutral 2: decrease set poing 1: Off (speed set disabled) The speed set point (target speed) shall be integrated over time. The time base depends on the time the lever is in the + or - position. I.e in the + position >1s=fast increase, <1s = slow increase in the - position >1s=slow decrease, >1s = fast decrease
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1167	The speed lever positions and the acknowledment pushbutton shall be evaluted by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1395	The positions of the speed lever and the acknowledment pushbutton shall be connected to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1168	It shall never be possible to select a speed reference that is greater than any speed limitation that is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1345	With the vehicle in speed mode and speed limitation applied, if the driver tries to increase the speed regardless the speed limit by means of the speed lever a warning message shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1169	Speed set reference is increased by three different increase times of the speed lever. 1. 1.If speed lever is set in '+' position → Speed set reference shall increase 1km/h per trigger. 2. If speed lever is set in '+' position for >500ms and <3s → Speed set reference shall increase 5km/h every 500ms. 3. If speed lever is set in '+' position for >5s → Speed set reference shall increase 5km/h fast, 0 to 360km/h in about 9 sec.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1170	Speed set reference is decreased by three different decrease times of the speed lever. 1. 1.If speed lever is set in '-' position → Speed set reference shall decrease 1km/h per trigger. 2. If speed lever is set in '-' position for >500ms and <3s → Speed set reference shall decrease 5km/h every 500ms. 3. If speed lever is set in '-' position for >5s → Speed set reference shall decrease 5km/h fast, 0 to 360km/h in about 9 sec.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1171	The speed set reference shall be shown on the IDU display.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1175	If speed set reference mode is activated the TCMS automatically regulates the speed to the requested speed reference.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1174	Speed set reference shall be automatically set to 0 if no traction is available and speed is v = 0 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.188
Description	A master controller for traction and braking effort demand shall be provided. The driver sets the demand of traction or braking power in different levels of controller positions.
Safety level	SIL 1

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1148	There shall be 2 encoders in the master controller with 6 binary outputs.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1176	The 6 binary outputs of each encoder shall be connected to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1573	In addition to the 6 binary output, each encoder shall provide: – 1 parity bit – 1 zero bit	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_09.03.Zefiro-Europe.CONCEPT.1574	TCMS shall supervise parity bit and zero bit of both the master controller's encoder. In case of non consistency signals maintenance event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1177	The traction/brake power reference shall increase linearly over the full movement of the master controller brake handle.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1178	The shall be a plausibility check between the two encoders and the status of the traction/brake trainlines.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1194	If a encoder is considered faulty event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1180	Both the requested and applied traction/brake power shall be shown on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

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ID	2F_09.03.Zefiro-Europe.TRS.370
Description	It shall be possible to perform a test of the Master controller from the IDU. In the test each position of the Master controller shall be tested by manually moving the Master controller based on instructions provided on the IDU.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1032	Master controller test shall be activated from the IDU only when logged in as a maintainer	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1033	The IDU shall present a screen that instructs the driver to put the Master controller in each position. For each position the test logic shall verify that the Master controller indicates the required position.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1031	**Master controller test is possible according to the following conditions:** Master controller test shall only be possible at stand still v=0. Master controller test shall set 100% service brake as long as the test is running.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1034	The Master controller test shall be possible to restart any time during the test sequence by pressing a button on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1035	The two encoders are tested during the Master controller Test If one of the two encoders is faulty the faulty encoder shall be deselected by the software and not used. If both encoders are faulty traction block shall be applied as long as the cab with the faulty Master controller is activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1036	**Master Controller handle test results:** Test done, both encoders are OK Encoder A faulty Encoder B faulty Master controller out of order	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1393	The result of the master Controller Test shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1394	An event shall be set detailing the result of the Master Controller test.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.361
Description	The master controller functionality shall be redundant.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1023	The Master controller is equipped with two redundant encoders and a set of trainlines to acquire the traction/braking effort information in both normal and emergency operation.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

In normal condition both encoders work, TCMS acquires and use the information from both encoders in order to provide a reliable traction/braking effort request.

ID	2F_09.03.Zefiro-Europe.TRS.371
Description	In case of Master Controller fault, Master controller back-up mode (sealed latched switch) can be selected. In this mode the driver can drive the train by using the Speed controller.
Safety level	N/A

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Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1181	If at any time, one encoder set differs more than two positions from the other, for a longer time than 5 seconds, an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1183	It shall always be the highest brake position or the lowest traction request from either contact set of the Master controller that is used to determine Master controller position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

If both encoders are faulty or there is a mechanical fault with the Master controller (Master controller Major failure) Traction block shall be applied till standstill and "back-up mode" enabled.

Urgency Brake will also applied till standstill in case of both encoders faulty according requirement 2F_05.02.Zefiro-Europe.CONCEPT.1055.

When the train is standstill the following instruction shall be displayed:
- Operate 'Master controller backup mode' switch
- Switch to Indirect Brake mode

The train need to be repaired soon after the end of service.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1188	If the "Master controller backup mode" switch has been operated AND Train Braking mode is set on "Indirect Brake" Then Master controller back up mode shall be considered ACTIVE	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1578	Traction block shall be applied if MCH backup mode is turned to backup position. Traction block shall be reset when – Train Braking mode is set on "Indirect Brake" OR – MCH backup mode switch in normal position OR – Emergency operation mode is active	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_09.03.Zefiro-Europe.CONCEPT.1541	If both encoders are faulty the Traction block shall be applied until the train is stopped and "back-up mode" is enabled: – Brake selector moved in "indirect brake" position – Back up mode switch moved in "enabled" position	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1533	"Master controller backup mode' switch can be operated only when the train is in standstill	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1185	If both encoders are faulty then an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1186	If the "Master controller backup mode' switch is operated then an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1536	An indication of the traction set point selected through speed lever shall be displayed on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1538	An indication that Master controller back up mode is enabled shall be displayed in IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1189	If the 'Master controller backup mode' is active then the speed lever shall be used to select traction reference only.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1190	When the speed lever is in the '+' position in Master controller back up mode then the traction set point shall increase linearly (till 100%) as long as the lever is kept in this position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1191	When the speed lever is in the '-' position in Master controller back up mode then the traction set point shall be reduced linearly (till 0%) as long as the lever is kept in this position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1192	If the speed lever is in the 'N' position Master controller back up mode then last traction setpoint shall be used as reference.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1537	When the speed lever is in the '0' position Master controller back up mode traction reference is set directly to 0%.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.209
Description	The ultimate position of the traction/brake shall cause emergency braking.
Safety level	SIL 4

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1028	**Last braking position of the Master Controller shall cause the application of the Urgency Brake.**	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 4
2F_09.03.Zefiro-Europe.CONCEPT.1195	An event shall be displayed on the IDU if there is an Urgency Brake application due to the Master Controller.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1282	If the Master controller is in Emergency brake position and Urgency brake is not applied within 2 seconds (Urgency brake loop is deenergised) then an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.1.4 Traction Safe

3.1.4.1 Function Design

The traction safe is a function to ensure that the train is ready to move before enabling the traction.
Traction shall be enabled only when all conditions for traction are fulfilled.
By energizing the traction safe train line, the traction will be enable.

To prevent the traction safe trainline shall be deenergized.

If positive traction is to be inhibited traction block shall be set (TCMS via MVB).
If positive traction and electrodynamic brake are inhibited, the “Traction safe” trainline shall be de-energized.

The traction safe can be in any case overridden via HW (All doors are not closed and locked from door, Low brake pipe pressure, Traction safe loop de-energised functions) and via SW (All doors are not closed and locked from doors, Low brake pipe pressure, External power supply is connected, External air supply is connected, Traction safe loop de-energised). Further description are included into the Traction Safe Override function described into this SFRS.

3.1.4.1.1 Normal condition

The traction safe trainline shall energized, and so traction shall be enabled only when all conditions for traction are fulfilled.

3.1.4.1.2 Failure cases

A) Traction is not released even if all the condition for releasing the function are satisfied.
B) Traction Safe trainline is energized even if same conditions are not satisfied.

3.1.4.1.3 Failure detection

A) Done by the driver (no traction provided with the master controller in traction position)
B) Alarm to be set on the IDU

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3.1.4.1.4 Failure handling

A) If traction is not released, override function can be used to clear the track. Safety procedure need to be defined to drive the vehicle with override function active. Train need to be put out of the service as soon as it is possible.
B) Stop the train as soon as possible. Towing/Restriction operation requested.

3.1.4.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.340
Description	Traction shall only be possible in the whole train when all conditions for traction release are fulfilled. For that case there shall be a hard-wired train line "traction safe".
Safety level	SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.373
Description	Traction shall be disabled if the pressure in the brake pipe is less then 3,5 bars.
Safety level	SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.389
Description	Use of the emergency brake, for whatever reason, shall automatically cut all traction power, without the facility of reapplying traction power while the emergency brake is being applied.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.828	Two cab to cab trainline for traction safe shall be provided.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.909	The traction is enabling by energizing a trainline	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1113	Traction safe trainline shall be de-energized when: – Green Loop (not all doors are closed) – No active cab – Brake pipe pressure less than 3,5 bars – Emergency Brake applied (Emergency Brake Loop De-energized) – TCMS DO are de-energized	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1585	it shall be possible to de-energize any traction safe trainline using TCMS DO	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1594	If TOWED mode is NOT active (from the active cab), TCMS shall de-energized both Traction safe trainline from the active cab (by means of safe DO) any time pressure in Brake Pipe drops below 350kPa. DO shall be reset with pressure riese over 400kPa Pressure signal for trigger shall be derived from second lowest valid TCG sensor in the train shall be used. (8 TCG sensor available in each train)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1396	Pressure switch installed in DM1/DM8 shall measure the pressure in the brake pipe and de-energise the traction safe loop if brake pipe pressure is less than 3.5 bars.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1401	If the brake pipe pressure switches are in different positions for more than 10 seconds an event shall be generated for the maintainer.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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ID	2F_09.03.Zefiro-Europe.TRS.334
Description	The traction demand shall be blocked to all motors in the train if any of following conditions is true: – master controller is in brake position – ATP service brake is active – too high speed in washing- or coupling mode OR – fault in speed calculation
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1052	Traction shall be blocked (traction block set) if any of the following conditions are set. (Traction block means that positive traction/brake reference is not allowed by TCMS) – Urgency brake is applied – Emergency brake is applied – No driver's desk is active – All Doors are not closed and locked from doors (including doors are released) – Master controller is in Brake Position – Master controller fault is detected – DBV in service brake position (if Direct Brake is enabled) – Low brake pipe pressure (brake SFRS concept.2174,1503) – Brake request from ATP is active – Inhibition traction is applied by the ATP – Passenger Emergency Brake is requested – The active cab is coupled – External power supply is connected at standstill – Traction safe loop is de-energised – No valid direction is set – Direction is changed at speed – Too low brake effort – Traction block request from washing speed mode – Traction block request from coupling speed mode – External air supply is connected (can be overridden from IDU) – Brake test is active – Traction test is active – Speed calculation fault – No traction is available – Emergency operation mode activated with master control handle in traction position – Change of driving mode with master control handle in traction position – No valid configuration – Decoupling sequence is active – Serious fault not acknowledged within 15sec – Change of direction with Master controller in Traction position – Traction block request from DIS system – ATP enabled in the not active cab (presence of life signal) – MCH backup mode switch active in DEP brake mode – - Speed limitation reset by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1539	To re-establish traction, once traction-block conditions have been reset, the requested effort shall be first set to 0 (Master controller shall be put in coasting position or traction setpoint shall be put to 0 when "master controller back-up mode" is enabled). This is valid for all conditions exept when traction block is determined by ATP system at speed. In this case effort will be re-established as soon as the block is removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1367	If the Traction safe trainline is considered to be energised the PCU shall be notified. Traction is considered to be enabled when below conditions is true: – both DI connected to a traction safe trainline are (1) on at least one trainline In case it is not possible to check conditon above (DI on a trainline are inchoerent or invalind) TCMS has to check that followin condition are true: – all doors (external and cab) in the train are closed and locked – the brake pipe pressure is above 4.0 bar – a cab is active in the train – Emergency Brake is not applied (Emergency Brake Loop energized) The status of the Emergency operation switch and the Doors interlock bypass switch shall be considered for above conditions to avoid false indication of events.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1283	If traction is considered to be enabled according requirement 2F_09.03.Zefiro-Europe.CONCEPT.1367 and the traction safe trainline status sent from the PCU indicates that any Traction safe input is de-energised then an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

TCMS shall check coherence between any of following elements and the related traction safe trainline:
- RBLE1/RBLE2 intervention in the active cab
- RBTS1/RTS2 intervention in the active cab
- PST intervention in the first two motor cars in the leading half train (half train with active cab)
- TCMS DOs for Traction safe in the active cab

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1587	If emergency operation is NOT active and Green Loop is Energized (or bypassed): Any time relay RBLE1 is de-energized (Dig Input, 1 → 0), TCMS shall immediatly check that related PST (DM car switch) status is still closed (Dig Input = 0) and DO on TL1 are still both closed (1), so after 0,1s (parameter to be tuned during commissioning) TCMS shall check that: – RTS1 relay status is de-energized (Dig Input 0 → 1) – Related Traction Safe Trainline (TL1) shall be de-energized (feedback DI = 0) In case RTS1 or TL1 are not read as de-energized an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1593	If emergency operation is NOT active and Green Loop is Energized (or bypassed): Any time relay RBLE2 is de-energized (Dig Input, 1 → 0), TCMS shall immediatly check that related PST (M car switch) status is still closed (Dig Input = 0) and DO on TL2 are still both closed (1), so after 0,1s (parameter to be tuned during commissioning) TCMS shall check that after: – RTS2 relay status is de-energized (Dig Input 0 → 1) – Related Traction Safe Trainline (TL2) shall be de-energized (feedback DI = 0) In case RTS2 or TL2 are not read as de-energized an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1588	Everytime emergency brake is released (RBLE1 energized in the active cab, Dig Input 0 → 1), TCMS shall check that as soon as TCG sensor in the DM car with active cab is above 450kPa, – Related PST switch shall be reset (Dig Input 1 → 0) – RTS1 shall be energized (Dig Input 1 → 0) within 0,5s otherwise an event shall be set The event shall not be triggered if MCB of RTS1 power supply is tripped.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1589	Everytime emergency brake is released (RBLE2 energized in the active cab, Dig Input 0 → 1), TCMS shall check that as soon as TCG sensor in the M car nearest to the active cab is above 350kPa, – Related PST switch shall be reset (Dig Input 1 → 0) – RTS2 shall be energized (Dig Input 1 → 0) within 0,5s otherwise an event shall be set The event shall not be triggered if MCB of RTS2 power supply is tripped.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1597	Everytime emergency brake is released (RBLE1 energized in the active cab, Dig Input 0 → 1), TCMS shall check that before PST pressure switchis in the DM car with active cab is trigger (1 → 0): – RTS1 is still de-energized (Dig Input 1) In case RST1 is energised (0) before PST is trigger an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1598	Everytime emergency brake is released (RBLE2 energized in the active cab, Dig Input 0 → 1), TCMS shall check that before PST pressure switch in the M car nearest to the active cab is trigger (1 → 0): – RTS2 is still de-energized (Dig Input 1) In case RST2 is energised (0) before PST is trigger an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1590	If Green Loop is energized (or bypassed) and RTS1 is energized (DI = 0), as soon as DO on traction safe line 1 (TL1) are energized (0→1), TCMS shall check that – TL1 status shall trigger (0→1) whithin 0,5s otherwise an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1591	If Green Loop is energized (or bypassed) and RTS2 is energized (DI = 0), as soon as DO on traction safe line 2 (TL2) are energized (0→1), TCMS shall check that – - TL2 status shall trigger (0→1) whithin 0,5s otherwise an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1592	If there are NO local active cab in all consist of the train (including when parking mode is active), TCMS shall check that both Traction Safe trainline are de-energized (all feedback DI = 0) otherwise an event shall be set (including detection of faulty trainline)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1347	If Emergency Brake loop is de-energized and the PCU indicates that any Traction safe input is energised an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1285	If any door closed relay (RGL1, RGL2) are indicating that at least one door is open and the green loop is not bypassed then if the any PCU indicates that a Traction safe input is energised an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1286	If the brake pipe pressure is below 2.3 bar and the PCU indicates that any Traction safe input is energised an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1115	If traction block is active then an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.827	One input for Traction safe shall be provided on the DCU/M to interrupt switching.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1287	The status of Traction safe input shall be forwarded to TCMS via the MVB.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.829	Any traction demand or brake demand shall be blocked if the traction safe input is deenergised. The traction safe input shall affect both traction and dynamic brake.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1417	If both Traction safe loop are de-energized then Traction block shall be applied.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1179	Traction block shall be set if no traction is available at standstill.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1575	Traction block shall be set in case ATP is enabled in the not active cab (presence of life signal)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1576	In case of Traction block applied due to ATP enabled in the not active cab, an advice shall be displayed to the driver asking to exclude ATP in the other cab	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.5 Traction Safe Trainlines Bypass

3.1.5.1 Function Design

Traction safe can be inhibited or overridden in case of emergency to clear the track. This action has to be confirmed actively by the driver.

Traction safe can be inhibited by means of two sealed latched switches:
1) can inhibit just the signal “Door interlocking bypass”;
2) can bypass the whole Traction safe loop (both trainlines) by means of an "emergency operation switch".

The pressure switch "PST" located in leading motor cars will be used to cut traction in case of EB.

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3.1.5.1.1 Normal condition

Traction safe shall be overridden in case of malfunction to clear the track and start traction.

3.1.5.1.2 Failure **cases**

A) Traction override is not working properly, and no traction is given
B) Unwanted Override

3.1.5.1.3 Failure detection

A) Done by the driver (even with the emergency operation active, traction is not available).
B) Message on the IDU.

3.1.5.1.4 Failure handling

A) Train has to be towed
B) Operational restriction and maintenance required

3.1.5.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.343
Description	It shall be possible for the driver to manually override Traction safe in case of malfunction to clear the track and start traction.
Safety level	SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.389
Description	Use of the emergency brake, for whatever reason, shall automatically cut all traction power, without the facility of reapplying traction power while the emergency brake is being applied.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1054	Traction safe can be inhibited or overridden in case of emergency by rotating the dedicated two latched switches on the driver desk: By means of the '**Doors interlocking bypass**' the following reason for traction block shall be overridden: – All doors are not closed and locked from doors By means of the '**Emergency Operation Switch**' the following reason for traction block shall be overridden: – All reasons except emergency brake (via pressure switch PST) that opens the traction safe loop and including Traction safe loop de-energised.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2

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2F_09.03.Zefiro-Europe.CONCEPT.1117	Traction safe can be inhibited or overridden in case of emergency by pushing the dedicated two latched switches on the driver desk: By means of the '**Doors interlocking bypass**' the following reason for traction block shall be overridden: – All doors are not closed and locked from doors By means of the '**Emergency Operation Switch**' the following reason for traction block shall be overridden: – All reasons except emergency brake (via pressure switch PST) that opens the traction safe loop and including Traction safe loop de-energised.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1118	Traction safe shall be inhibited or overridden in case of emergency by pushing the dedicated two latched switches on the driver desk: By means of the '**Doors interlocking bypass**' the following reason for traction block shall be overridden: – All doors are not closed and locked By means of the '**Emergency Operation Switch**' the following reason for traction block shall be overridden: – All reasons including Traction safe loop de-energised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1055	The interruption of the door interlocking trainline and traction safe bypass shall be inhibited temporarily as long as a switch in the cab is activated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1056	A latched switches to bypass the door interlocking trainline shall be provided in the cab.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1057	The operation of the door interlocking bypass function shall be supervised by TCMS, generating an event if the button in cab for door interlocking bypass is activated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.845	The interruption of the traction safe trainline shall be inhibited temporarily as long as a button or switch in the cab is activated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.846	A latched switches to bypass the traction safe trainline shall be provided in the cab.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2

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3.1.6 Emergency Operation mode

3.1.6.1 Function Design

In case of emergency, for example for fire incident or failure of electrical equipment, the train can be operated in emergency operation mode by selecting the "emergency operation mode" sealed switch.

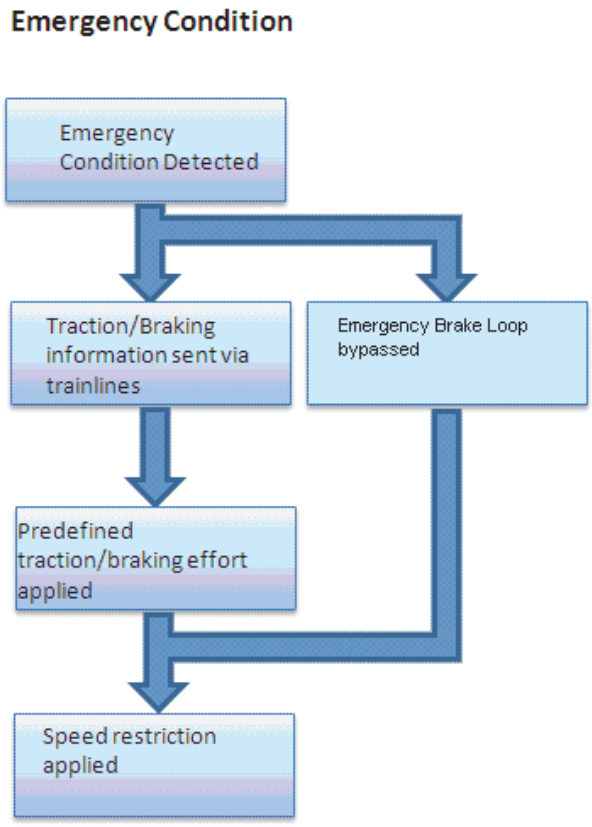
A dedicated trainline is provided to inform all the involved system.

Conditions that involves Emergency Operation:
- TCMS cannot read the controls in the cab (e.g. a fire incident that isolate MVB connection)
- Electrical failure involving Emergency Brake loop (e.g. trailine interruption, failure on equipment commanding EB via electrical loop)

When the "Emergency Operation" mode is active:
- the traction/brake information will be distributed by means of 2 fire resistant trainlines that connect directly the master controller with all the BCUs. The demand will be forwarded by the BCUs to the propulsion system via the main MVB line
- the emergency brake loop will be bypassed (energizing directly the two EVAPP valves by means of the Emergency Operation trainline.

The safe application of an Emergency brake is guaranteed by the direct pneumatic venting of the Brake pipe via EB Pushbuttons and DBV, instead in case of request from Master controller the EVAPP valves will be de-energized.

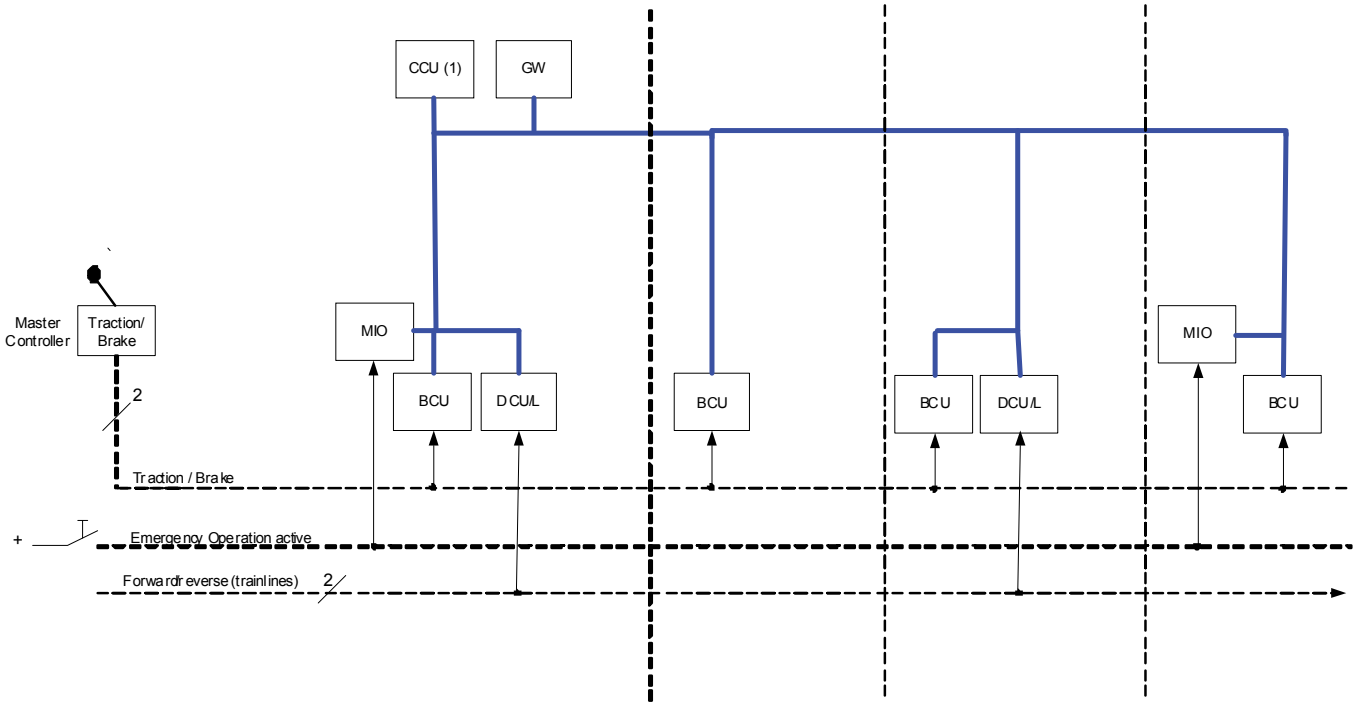
- The traction effort required is pre-defined in order to keep a train speed of 80km/h



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Architecture of emergency traction/brake trainlines



3.1.6.1.1 Normal condition

The emergency operation mode shall be activated anytime by the driver using a sealed switch. The manual activation shall ensure that the driver is aware about the risk and the limitation that this mode brings. In addition, the speed limit set by TCMS should avoid dangerous situations due to high speed.
In emergency operation mode, the traction and braking demand from the brake handle to TCMS may not be available.
And in order not to deprive the train of more than 50% of its traction power, the traction effort will be distributed by trainlines only and read locally by the propulsion control.
If the emergency operation mode' is activated, then the 'emergency operation mode' trainline is energised. The status of the 'emergency operation mode' trainline shall be monitored by TCMS in every 4 car segment.
If the 'emergency operation mode' trainline is set then a predefined amount of traction and brake demand shall be demanded of the BCU and MCMs depending on if the traction or brake trainlines are energised. The traction and brake trainlines are read by the BCU in each car.
In addition, in case of failure to the emergency brake loop the "emergency mode" trainline is used to keep EVAPP valves (that are used to vent the brake pipe) energized allowing the train to move and escape from dangerous situations.
'Forward' and 'reverse' trainlines functionality shall not be affected.

3.1.6.1.2 Failure cases

A) No Traction/Braking effort is manageable through the trainlines
B) Some traction/braking effort signals are not available
C) Emergency Brake Loop cannot be bypassed

3.1.6.1.3 Failure detection

A) Done by the driver (no possibility to move the train)
B) Done by the driver (some master controller positions do not provide any response)
C) Done by the driver (train cannot move and Brake pipe pressure is 0bar)

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3.1.6.1.4 Failure handling

A) Tow is needed
B) Proceed with the available efforts
C) Bypass EVAPP valves manually (using RIVAPP cocks)

3.1.6.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.337
Description	The distribution of traction demand level shall be ensured to at least 50% of all DCUM along the train for minimum 15 minutes in case of fire.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.812	If Emergency operation mode is active and the Emergency traction demand input is set then the traction demand shall be regulated to keep a constant train speed of 80km/h. The traction demand sent from the Master controller shall be ignored.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.196
Description	The driver shall confirm the emergency operation by a dedicated button or switch. Emergency operation shall be active only if this switch is activated by the driver.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1046	There shall be a 'emergency operation mode' sealed switch on the drivers desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1120	The 'emergency operation mode' sealed switch on the drivers energises the 'emergency operation mode trainline'.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1119	The status of the 'Emergency operation mode' trainline shall be connected to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1121	The status of the 'Emergency operation mode' sealed switch shall be connected to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1047	When the 'emergency operation mode' sealed switch is pressed there shall be an event logged.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1449	When "emergency operation" mode is active, and indication shall be displayed on IDU. In addition the prescribed speed limitation has to be shown.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1122	If there is a discrepency between the 'emergency operation mode' sealed switch and the 'emergency operation mode' trainline then an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.231
Description	In case of an incident, the activation of braking operation shall not cause braking down to stand still. Any evacuation drive must be ensured with minmum time delay.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.524	In case of any brake application due to fault the brake application shall release after switch over to emergency operation when MCH is moved into traction position.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1410	If MCH is in traction position (traction trainline energized before emergency operation trainline) then traction shall be blocked until traction trainline is reset.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.192
Description	The emergency operation mode is used as backup for traction and brake control in case of a major incident like fire. The redundancy of the normal communication paths via MVB/WTB is not enough because no fire safe communication bus cables are available on the market.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.194
Description	The master controller shall be connected directly to this backup communication path though being independent. In emergency operation only the traction/brake controller and the indirect brake lever (DBV) can be used. The emergency operation mode shall allow the train to run even in case of fault on the Emergency Brake Loop.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.394	The master controller shall provide a separate contact set to be connected to **2** trainlines.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1049	The 'traction trainline' and 'brake trainline' shall be set by the traction and brake areas of the Master controller.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1124	The 'traction trainline' and 'brake trainline' shall be supervised by TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1126	If the 'emergency operation mode trailine' is energised then the TCMS will react according to the change of status of the trainline.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.656	The traction demand also in Emergency Mode operation shall be sent to propulsion control via the local car MVB. (reason: propulsion is anyway dependent on that MVB)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1611	When Emergency Operation mode is activated with Doors released, if communication with driver's desk is ok, once traction is demanded an advisory shall be set to the driver indicating that a closing command of the doors shall be issued.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1420	When traction trainline is set during emergency operation mode then HV shall be reactivated if HV was active when emergency mode was activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1609	In case of automatic re-configuration of HV due to missing communication with driver's desk, when the driver puts the MCH in braking position for a time less than 2 seconds and then move it to coast TCMS shall open the LCB and lower the pantograph. When MCH is moved again to traction position TCMS shall re-configures automatically HV according previous configuration.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_09.03.Zefiro-Europe.CONCEPT.1610	In case driver's desk key position, LCB switch status, pantograph selectors or line voltage selector cannot be detected or redundant TCMS feedbacks are incoherent an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1447	If emergency operation mode is activate, the Emergency Brake loop shall be bypassed energizing directly EVAPP valves.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3
2F_09.03.Zefiro-Europe.CONCEPT.1448	If an Emergency Brake is commanded via Master Controller (MCH), during emergency operation mode, the EVAPP valves bypass shall be released.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 3

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Emergency Operation shall be able to move the train, also in case of multiple fire alarms set in technical areas of both half trains.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1599	In case of Fire alarm detected in technical areas, if Emergency Operation is activated, following actions shall be granted: – all pantographs shall be available to the driver, except for those ones previously faulty (not in case of ADD intervenction by-pass) OR cut-out; – LCB of the raised pantograph shall be allowed to be closed; – ACMs of the half train with pantograph raised shall be allowed to starts; – MCMs of the half train with pantograph raised shall be allowed to starts; **Note:** 220V sockets and MV auxiliary contactors (Gallley in DM1, Bistro in M3) shall be kept deactivated. Also, separation contaction of MV (contactor 6) and HV root line disconnector have to kept open.	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.7 Train Speed Calculation

3.1.7.1 Design

Purpose of this function is to describe how to calculate train speed. Train speed is calculated by means of the available speed sensors belonging to the brake system and the propulsion system.
All speed sensors correctly working will be used for the train speed calculation but for the following function:

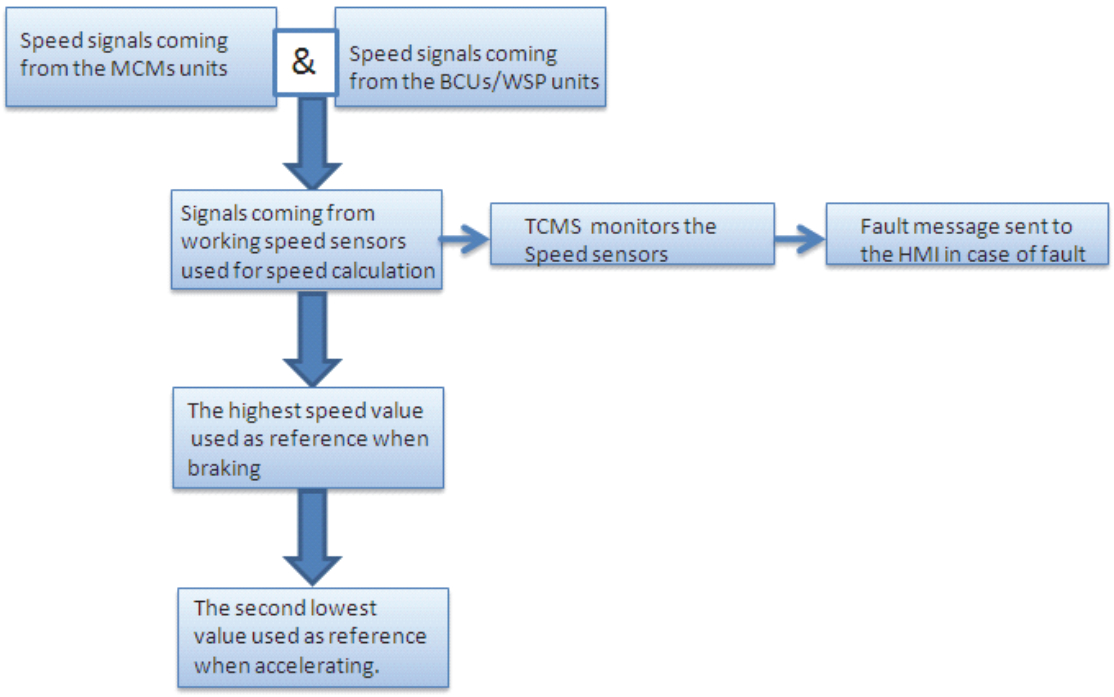
– doors interlocking:
BCU/WSP will be used to provide the reference speed.

– visualization to the Driver in case ATP cut-off:
BCU/WSP speed will be displayed in a safe way
(optional DIS will be used - non safe)

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Speed Calculation:



3.1.7.1.1 Normal condition

There is consistency between the different speeds signals coming from the 8 car unit used for train speed calculation.

3.1.7.1.2 Failure cases

a) Speed signal of one or more sensor/system are not align to train speed but at least 1/3 of the speed sensors in the 8 car unit used for train speed calculation are correctly working.
b) Less than 1/3 of the speed sensors in the 8 car unit used for train speed calculation are correctly working.

3.1.7.1.3 Failure detection

a) IDU → Message sent to identify where are located the faulty speed sensor
b) TCMS → Major alarm activated on the IDU & Diagnostic A-event buzzer to be activated

3.1.7.1.4 Failure handling

a) TCMS Excludes the faulty speed sensor/s from reference speed. Maintenance needed at the end of the service.
b) Try a vehicle master reset, if the faults are still present then proceed to the closest station and retire the vehicle from the service. Maintenance needed.

3.1.7.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.199
Description	The speed signals from all available DCUM and WSP on one MVB shall be used for train speed calculation.
Safety level	SIL 2

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Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.973	Each BCU/WSP (per car) and MCMs will send the speed signal per axle to TCMS and TCMS shall calculate the train speed based on valid speed sensors.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1130	MCMs shall send the speed signal per axle to the TCMS based on valid speed sensors.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1131	Each BCU/WSP shall send the speed signal per axle to the TCMS based on valid speed sensors.	Derived Requirement	4S_07.-.Braking System		SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.158
Description	The speed sensors from brake- and propulsion system in a 8 car unit will be used to calculate the train speed. The strategy of the calcualtion is to use the hightest speed value when braking and coasting and the second lowest value when accelerating.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.974	TCMS shall calculate the speed on a 8 car unit basis to ensure fast signal sampling times. The speed sensor from the first 8 car unit (considering as first car the active cab both in single train and in multiple operation) will be used to calculate the train speed according to the following strategy: – The highest speed value shall be used as reference value when braking, – The second highest value of trailer cars shall be used as reference value when not at braking. – If no valid trailer car speed is present then the lowest speed value shall be used when not at braking.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1414	If no valid speed can be calculated traction block shall be applied.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.161
Description	The speed sensors shall be continually supervised and faulty sensors shall be excluded from the speed calculation
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.975	The speed signals shall be continually supervised. An event shall be sent and presented on the IDU which speed sensor is faulty. Faulty speed signals shall be excluded for generating train speed.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.962	When the difference between max and min speed is > 6km/h an event shall be generated. The comparison shall not be active during slide.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.963	For axles with two speed sensors (motor axles) the two speeds shall be compared and both be excluded from the speed calculation if the speeds differs more than max of (1% of train speed and 1km/h). The comparison shall not be active during slide.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.399	The axle speed from all DCUM axles shall be provided to TCMS.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.400	The axle speed from all BCU/WSP axles shall be provided to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.860	TCMS shall send the train speed to propulsion control for slip regulation. The signal shall be calibrated considering the wheel diameters.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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ID	2F_09.03.Zefiro-Europe.TRS.200
Description	The speed calulation for – visualisation in the cab – doors interlocking – DSD function shall be provided by ATP or other SIL2 computer, e.g. the Brake Controller BCU or WSP. In case of ATP cut out : – TCMS shall calculate the speed by means of the available BCU. – The speed provided by DIS system shall be used for visualization in the cab
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1450	If the ATP speed signal is not available (ATP cut off) the speed from the BCU/WSP shall be displayed on the IDU. (optionally DIS speed signal will be used)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

3.1.8 Speed Limitation

3.1.8.1 Design

3.1.8.1.1 Normal condition

The train speed shall be limited depending on the train status.

3.1.8.1.2 Failure cases

1) Speed limitation is not applied when a restriction is required
2) Speed Limitation applied when not requested

3.1.8.1.3 Failure detection

1) Detected by driver
2) Detected by the driver by means of message on the IDU

3.1.8.1.4 Failure handling

1) Speed to be regulated by driver
2) Try a vehicle master reset, if the faults are still present then proceed to the closest station and retire the vehicle from the service. Maintenance needed.

3.1.8.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.223
Description	If ATP is cut out, the speed shall be limited to 90km/h (TBD).
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.431	Traction block shall be applied when DIS request traction block. (For example ATP cut out)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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ID	2F_09.03.Zefiro-Europe.TRS.155
Description	Depending on the status of the train speed limits are applied. The most restrictive speed limit shall be used. The following speed limits are defined: – driving without ATP (ATP cut out) – coupling speed – washing speed – IMS warning (manual speed reduction by driver) – IMS alarm (manual speed reduction by driver) – one or more WSP cut out or not working – one or more BCU cut out or not working (EP brake) – overheat in propulsion system – overheat in line voltage system – limited power from catenary – pantograph error (DC current limitation)
Safety level	N/A

The speed limitations will be enforced by a combination of
- an automatic speed restriction by TCMS with no SIL and
- a manual restriction done by the drivers following operational rules to achieve SIL2

The driver will notice that a speed limitation is in force:
1. ATP isolated: The dedicated lamp (LCEA) on the driver desk is switched on and both ATP DMI are black (Traction block requested by DIS)
2. coupling speed will be selected actively by the driver from standstill
3. washing speed will be selected actively by the driver from standstill
4. speed limits reated to WSB/BCU cut out or error will be indicated to the driver in a safe way (serious fault), details on the IDU.
5. speed reductions due to IMS uses trainlines and alarm lamps in the drivers desk is described in the bogies SFRS.
6. overheat or power limitations related to propulsion and high voltage will be detected internally in the propulsion system and the torque will be limited. I.e. the system protects itself without driver interaction.
7. Pantograph error: see the line voltage SFRS.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.236	The speed limit with the lowest (most restrictive) value shall always be used to limit the speed automatically by reducing the traction demand when the speed limit is reached. The traction demand shall be ramped down to zero at a speed of 10% beyond the speed limit. This shall ensure that the train still can drive with fulload at an inclination.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1451	The active speed limit shall be displayed on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1579	Whenever a speed limitation is reset by TCMS, a traction block shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.9 Speed Calibration

3.1.9.1 Design

All axle speeds used for Speed Calculation shall be calibrated using an independent speed source to determine the true speed.

3.1.9.1.1 Normal condition

The calibration procedure shall start upon manual request from the driver and shall use ATP speed as reference. The train shall be coasting to enable the calibration. The diameters calculated for all speed signals received by TCMS shall be stored in TCMS.

3.1.9.1.2 Failure cases

A) Calibration fails to initiate when calibration required
B) Calibration initiates but no update of wheel diameter performed

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3.1.9.1.3 Failure detection

A) Uneven regulation of speed and uneven ride
B) Uneven regulation of speed and uneven ride

3.1.9.1.4 Failure handling

A) Maintenance needed before the train is entered into traffic after a wheel set change
B) Maintenance needed before the train is entered into traffic after a wheel set change

3.1.9.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.154
Description	The axle speed singals from the BCU/WSP and DCUM shall represent a rotational speed only.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.407	The DCUM shall provide speed signals for all motor axles to the TCMS.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.408	The BCU/WSP controllers shall provide speed signals for all axles to TCMS.	Derived Requirement	4S_07.-.Braking System		SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.201
Description	The wheel diameters shall be stored in the TCMS for calibration of the speed signals received from BCU/WSP and DCUM (rotational frequencies). The calibration shall be done using the speed from the ATP system.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1368	The speed from ATP shall be sent to all PCU's in the train to be used for wheel diameter calibration, only if the corection factor for all BCU is refering to 920 mm. If it is not the case during the calibration the speed signal shall be the TCMS signal	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.554	The wheel diameters shall be calibrated towards the reference speed when command for wheel diameter calibration is set from TCMS.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1400	The signal provided by the ATP shall be used to verify the train speed measured by TCMS.	Derived Requirement	4S_09.06.02.-.National Safety System		SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.212
Description	Speed calibration shall ensure the consideration of wheel diameters in the train speed signal calculation.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.203
Description	The calibration procedure shall only be activated manually from a driver or maintainer after a workshop visit affecting the wheel diameters.
Safety level	SIL 0

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Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.404	A speed calibration request shall be activated if the condition belowe are soddisfy, between a speed range of 100-300 km/h: 1 - Calculated Train Speed from TCMS, derived from BCU/DCUM, differs more than 3km/h from ATP speed, when train is in coasting or in traction and no slip/slide is active from BCU/DCUM with speed higher than 100 km/h for more than 10 seconds. TCMS shoud store it and if the condition 4 to 9 are valid a calibration requerst has to be performed. This condition has to be reset if the calibration is passed **OR** 2 - The difference between max and min speed read by all the sources is > 6km/h, when train is in coasting or in traction and no slip/slide is active from BCU/DCUM with speed higher than 100 km/h for more than 10 seconds. TCMS shoud store it and if the condition 4 to 9 are valid a calibration requerst has to be performed. This condition has to be reset if the calibration is passed **OR** 3 - If one speed of the DCUM is geater than 3 km/h compared with the train speed. This criteria is valid only if train speed is more than 100 km/h for 10 seconds and no slip/slide is active from the PCU. If this condition ia active the TCMS shoud store it and if the condition 4 to 9 are valid a calibration requerst has to be performed. This condition has to be reset if the calibration is passed If condition 1 **OR** 2 **OR** 3 is active, the out put has to be put in **AND** with the conditions below 4 - Pneumatic brake not applied for more than 5 seconds **AND** 5 - Traction/brake effort reference is between 0 and +20% for more than 5 seconds **AND** 6 - One LCB closed in the train (this condition has to be the same in multiple conposition, one LCB closed per train) **AND** 7 - ATP Acceleration/Decceleration < 0.10 m/s^2 **AND** 8 - The speed range is between 100-300 km/h, the ATP signal has to valid **AND** 9 - No slip/slide is active for more than 5 seconds If a speed calibration is activated, an event should be set in Telediagnostica, there should be also an event in Telediagnostica that explaince why the calibration is requested, (condirion 1, 2 or 3). The duration of the calibration should be at least 15 seconds The TCMS should activate also an event in Telediagnostica if the calibration is passed or not • The calibration is passed if the condition from 4 to 9 have been present for at least 15 seconds • The calibration is failed if the condition from 4 to 9 have been present for less than 15 seconds. The event should explain why the calibration has failed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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2F_09.03.Zefiro-Europe.CONCEPT.88	During speed calibration the BCU speeds shall be calibrated towards the ATP reference speed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.405	If ATP speed signal is not available the speed calibration cannot be initiated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.406	An event shall be set in case a speed calibration request is activated specifying which is the cause that has triggered the request	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.411	The button to trigger the speed calibration on IDU should not be foreseen	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

3.1.10 Distance Counter

3.1.10.1 Design

The travelled distance function is an internal TCMS software function.

3.1.10.1.1 Normal condition

A counter shall increase by 1 every 1 kilometer and stored in non volatile memory

3.1.10.1.2 Failure cases

Counter not increasing when train has been in traffic

3.1.10.1.3 Failure detection

No increase in kilometers travelled from last upload of event data

3.1.10.1.4 Failure handling

Maintenance needed

3.1.10.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.163
Description	An odometer function logging the travelled distance shall be provided for maintenance use. The data shall be stored in non volatile memory and be available for the Telediagnostica application.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.94	An odometer function shall logg the travelled distance and store it in battery backed up memory. The data shall be available for the Telediagnostica application.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.93	The resolution of the Odometer function shall be 1km	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.11 Master Reset

3.1.11.1 Design

Master reset is a TCMS software controlled reset of certain permanent failures and blockings.
Different systems use interlocks for blocking of faulty equipment. The fault interlock must have a reset function with unlimited number of available resets that is available for the maintenance/service people when train is in Maintenance mode only.

3.1.11.1.1 Normal condition

The master reset can be performed by the maintenance personnel during maintenance or by the driver during operation from the IDU. Master reset is allowed by the TCMS just at standstill condition (v=0).

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3.1.11.1.2 Failure cases

Master reset is demanded but no reset is performed.

3.1.11.1.3 Failure detection

Faulty equipment not reset and continue to influence train performance

3.1.11.1.4 Failure handling

Not known (Manually disconnect/reconnect the power supply of the faulty unit to force the faulty system to be re-started in order to verify if in this way the permanent block can be eliminated.)

3.1.11.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.165
Description	Different systems use interlocks for blocking the use of faulty equipment. That interlock has a reset function that is available for the driver/maintener. The reset is called “Master Reset”.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.99	A button on IDU shall be provided to trigger a "master reset".	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

ID	2F_09.03.Zefiro-Europe.TRS.213
Description	A "Reset" function shall be provided for the maintenance crew to reset applied interlocks of faulty equipment. The reset accessible to maintainers shall be unlimited but be protected against unauthorised access.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.432	A "master reset" shall be sent to all subsystems when following conditions are met: – all pantograph are lowered; – no external power supply is connected;	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1613	A dedicated event shall be set to the Driver when Master reset PB is pressed but any of the following conditions is not met: – all pantograph are lowered; – no external power supply is connected;	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.238	The "Master Reset" signal shall be received from IDU indipendently from which mode is active (Driver, Maintenance, Crew)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1135	The "Master Reset" function can be unlimited requested by Driver and Maintainer in the active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1390	The "Master Reset" function is allowed by the TCMS only at standstill condition (v=0).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.605	An event shall be logged if "master reset" has been trigged	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

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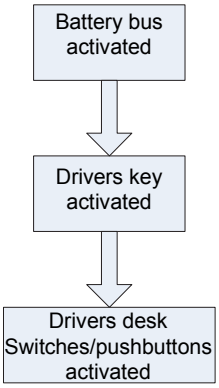
3.1.12 Drivers Desk supervision

3.1.12.1 Design

If the cab is activated (occupied), the indicators and control elements are active.
They will be hardwired activated by switch/relay when battery bus and drivers key is activated.

Lamp test uses a local pushbutton. All control lamps, gauges in control of hardware or software are tested. The hardwired lamps will be tested hardwired.

Block diagram Drivers desk activation ½ trainset



3.1.12.1.1 Normal condition

The driver's desk indicators and controls are available when the cab is activated.
A lamp test function is available by pressing a pushbutton on the drivers desk.

3.1.12.1.2 Failure cases

1. Specific button, lamp or display not working
2. Lamp test is not working

3.1.12.1.3 Failure detection

1. Driver observation
2. Driver observation

3.1.12.1.4 Failure handling

1. Depends on indication or control button failure
2. Depends on indication or control button failure

3.1.12.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.168
Description	Control from driver desk buttons and switches shall only be active in the active cab (excluding the emergency stop pushbutton and the driver key which are active in all cabs).
Safety level	SIL 2

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Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.104	All control elements in the cab besides the two emergency stop push buttons, the battery on push button, the parking mode pushbutton and the drivers key shall be disabled in the non-active cab. The supply to those controls shall be disconnected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.608	All control elements in the cab besides the two emergency stop push buttons, the battery on push button, the parking mode pushbutton and the drivers key shall be disabled in the non-active cab.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1577	Events shall be set for main driver actions on driver desk (eg LCB command, panto, doors, key, handles, bypass switches etc..)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.169
Description	Indicators, lamps and buzzers controlled by hardware directly, shall be disabled in the non-active cab a part from the paking mode lamp and door release indication.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.245	Indicators, lamps and buzzers controlled by TCMS software shall be disabled in the non-active cab a part from the paking mode lamp.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.434	Indicators, lamps and buzzers controlled by hardware directly, shall be disabled in the non-active cab a part from the paking mode lamp and door release indication.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.171
Description	Test of hardware controlled lamps, indicators shall be done upon manual command to ensure the correct functionality. (SIL 2 by means of the perfomance of the manual test by the driver -> driver´s manual)
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.436	A lamp test button shall be provided on the drivers desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_09.03.Zefiro-Europe.CONCEPT.501	The lamp test for hardware controlled lamps shall be triggered directly from the lamp test button. The lamp test button shall also be connected to TCMS digital input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.377	Lamp test of hardware controlled indicators (gauges, lamps) shall be done by manual command from the activated desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1136	If the lamp test button is pressed, all indicators shall be tested: – the lamps shall be completely lit	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.214
Description	Test of TCMS controlled lamps, indicators shall be done upon command to ensure the correct functionality.
Safety level	SIL 0

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Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.251	Lamp test of TCMS controlled indicators (gauges, lamps) shall be done by manual command from the activated desk. If the lamp test button is pressed, all indicators shall be tested: – the lamps shall be completely lit – the IDU (the primary and the secondary) shall show a white screen testing all pixles.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.502	When the lamp test button is pressed all indicators shall be activated simultaneously, the software controlled and the hardware controlled buttons. The synchronisation of all indicators shall be ensured.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.13 Serious Fault Indication

3.1.13.1 Design

A "serious fault" shall be notified to the driver via a dedicated indication. If a fault occurs that needs to be monitored or that implies a speed reduction, this lamp shall be lit.
When an event occurs, the driver shall acknowledge it, otherwise Traction block will be set until the acknowledge is done.

There are different types of fault that can be classified as "serious":

C1) Events that need the driver to stop the train in the next safest location (areas that have low fire risk and that can easily allow rescue operation).
The lamp will be steady lit also after the acknowledge. When the train is stopped the driver can check events and operational instruction eventually solve the failure.
The indication will be reset only after the failure is solved.

C2) Events that require a speed reduction (with a pre-defined speed limit) and that shall be solved within a time amount [ΔTmax] via unscheduled maintenance operations.
The lamp switches off only when the speed is under the prescribed limit and it will be lit again if this limit is exceeded.

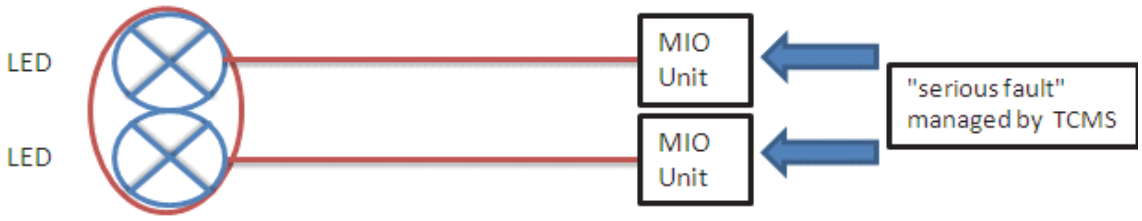
C3) Events that don't have any speed constraints and that can be simply acknowledged by the driver. In any case maintenance check is needed within a fixed period [ΔTmax] to solve the failure.
Class C3 failures will be displayed only as soon as the train is in standstill with Doors released (there isn't any immediate visible indication when the event occurs).
The lamp is switched off after the acknowledge by the driver.

When another desk is enabled the indication will be switched on again and the driver shall recognize the fault before start (if possible)
The value of ΔTmax should be no more than 24hrs.
The "serious fault" indication can be reset only after that the proper actions are performed (by the driver of by maintenance operators).
In any case if the maximum amount of time is elapsed and the serious fault is still active the lamp will be lit again and TCMS will not allow the train to start.
If this occurs during the normal service operations next time the train is stopped with released doors it will be not allowed to move again.
Alarms or faults that involve an automatic reaction from the train (eg emergency brake application) are not classified as "serious faults".
A dedicated redundant LED shall be installed on the drivers desk.

The Serious fault indication lamp shall be directly controlled by the TCMS via digital output. Because of this the lamp behaviour can be tuned according to the requests of the different authority.

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3.1.13.1.1 Normal condition

In normal condition both LEDS shall be activated after TCMS identify serious fault.

3.1.13.1.2 Failure cases

1) Only one of the two LEDs is activated after major fault identification.
2) None of the controls is activated after major fault identification.
3) One or both indicators are active even if there is not fault.

3.1.13.1.3 Failure detection

1) Done by the driver.
2) Done by the driver due to the additional information on the IDU.
3) Mute the Buzzer by means of the dedicated acknowledge button and/or ignore the Lamp.

3.1.13.1.4 Failure handling

1) During the maintenance done to eliminate the major fault, also the LED needs to be repaired.
2) During the maintenance done to eliminate the major fault, also the LEDs need to be repaired.
3) Maintenance is needed at the end of the service.

3.1.13.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.226
Description	A "serious fault" shall be indicated to the driver via a dedicated lamp on the drivers desk.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1085	A redundant lamp with two LEDs shall be provided on the driver desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1142	A "serious fault" redundant LEDs shall be connected to TCMS output. The two LEDs shall be connected to 4 different MIO-S outputs on two independent MIO-S units.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

ID	2F_09.03.Zefiro-Europe.TRS.376
Description	TCMS shall illuminate the "Serious fault" lamp on the driver's desk if a 'serious fault' has occured.
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1515	When a "serious fault" occurs the driver shall acknowledge it via hardkey on IDU. If no ack is detected within 15 seconds Tracion block will be set until the acknowledge is done.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1516	Serious fault alarm can be acknowledged using the hard key on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

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2F_09.03.Zefiro-Europe.CONCEPT.1550	When new "serious fault" is present the lamp on the driver's desk shall be lit and the buzzer in the cab shall sound. The buzzer shall be muted when the ack is pressed	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1079	The lamp shall extinguish if conditions for rectification are fulfilled, i.e. the fault is bypassed, or fault condition is inactive.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1540	If a second serious fault occurs when the lamp is already switched on, the buzzed shall start to sound and the new fault shall be acknowledged by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1519	When a serious fault is active and a new cab is activated, the lamp shall be switched on in the new cab and the fault needs to be again acknowledged by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1581	Serious fault supervision shall be enabled after train start up as soon as following conditions are met: – One cab activated in train AND Pressure in Main reservoir is greater than 700kPa (from at least one BCU in each consist of the composition, for multiple unit operations) OR – Cab is active for 10 minutes OR – Cab is active in train AND train is moving In case Automatic Brake test is active Serious fault supervision shall be disabled ex	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1522	Following events shall be considered as Serious Faults (according classes C1, C2, C3) C1: – EB application valve and pressure switch supervision – Dragging Brake (undue brake application) – BCU and WSP fault (or missing comunication) in a single car – Triple Stage Supervision (undue stage application) C2: – DNRA function loss in one or more cars (DNRA disturbed) C3: – Brake pipe neutral and isolation valve (EVNEU/EVISO) supervision – Triple Stage Supervision (missing stage application)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1526	C1 class faults shall lit the lamp as long as proper actions are not performed and failure is not solved	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

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2F_09.03.Zefiro-Europe.CONCEPT.1595	Following condition shall be used to "isolated" related Serious Faults (C1 faults) once triggered BCU and WSP fault → RIEP/DIST cock on isolated position in the affected car Dragging brake → n/a (trigger condition will be reset automatically after driver isolation action) Triple Stage Supervision (undue stage application) → RIEP/DIST cock on isolated position in the affected car (in this case also the "missing stage supervision" fault (C3) will be isolated) EB application valve and pressure switch supervision (EVAPP) supervision → RIVAPP cock on isolated position for the affected valve	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1527	C2 class faults shall lit the lamp as long as the train is over the prescribed speed limit	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1525	In case of C2 faults the related speed limit shall be displayed on IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1572	New C3 class faults shall be displayed (lamp is switched-on, alarm is shown and acknowledge is required) only when the train comes to standstill and doors are released	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_09.03.Zefiro-Europe.CONCEPT.1528	For C3 class faults the lamp shall be switched off soon after the driver acknowledges the alarm	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1517	A time counter shall be set as soon as an event classified as serious fault occurs (only for failures belonging to categories C2 and C3). The value of the counter [ΔTmax] shall be settable indipendently for each fault	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1521	The counter shall be reset as soon as the serious fault has been solved.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1518	If a counter is elapsed an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1424	The serious fault can be acknowledged once to temporarily continue driving. After the timer is elasped, next time the train is standing with released doors and the serious fault is still active, an alarm shall be displayed on IDU stating that the maintenance time is elapsed and it is not safe to drive the train.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1425	The reason for the serious fault shall be stored in NVRAM for Diagnostic.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1426	An event shall be set when serious fault acknowledge button is pressed.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1427	The serious fault lamp shall be part of the lamp test.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1520	The serious fault lamp shall be part of the lamp test.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

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3.1.14 TCMS configuration waiting indication

3.1.14.1 Design

Any change in the train configuration needs to be notified to the driver to avoid that wrong operation are performed while the Train Logic is busy.
A lamp controlled by TCMS will be installed on the driver's desk. The activation of this lamp is done in such a way that if there is any problem and the TCMS is not responding the indication is lit.
The lamp shall be steady lit if:

- a test of HV/Propulsion equipment is in progress
- HV disconnectors reconfiguration
- selected HV configuration is not in line with the catenary voltage
- TCMS is not responding and/or faulty

The lamp is flashing in multiple train operation if:

- TCN is faulty
- Bus Inauguration is not successful

3.1.14.1.1 Normal conditions

In normal condition the lamp should be off.
If the lamp is lit steady or flashing the driver shall wait for the TCMS to complete the ongoing operations.

3.1.14.1.2 Failure cases

1) Lamp is not lit when ordered
2) Lamp is lit also if there isn't any fault

3.1.14.1.3 Failure detection

1) Done by the driver during lamp test
2) Done by the driver checking detailed information on IDU

3.1.14.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.382
Description	One lamp indicating that the train configuration is changing shall be installed on Driver's Desk
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1487	One lamp indicating that the train configuration is in progress shall be installed on Driver's Desk	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1488	The lamp shall be connected to a TCMS Digital Output in such a way that if the DO is low the lamp will be switched on	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1497	The lamp is connected to a TCMS Digital Output in such a way that if the DO is low the lamp is switched on The lamp shall be steadly lit if: – a test of HV/Propulsion equipment is in progress – HV disconnectors reconfiguration – selected HV configuration is not in line with the catenary voltage – TCMS is not responding and/or faulty The lamp is flashing in multiple train operation if: – TCN is faulty – Bus Inauguration is not succesful	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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2F_09.03.Zefiro-Europe.CONCEPT.1508	If the train is waiting for the new configuration and the lamp is lit (steady or flashing) the software traction block shall be applied (if possible) and traction effort shall be set to zero	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.15 Cab doors supervision

3.1.15.1 Design

The cab doors will be equipped with two double contacts switch for indicating the status of the cab door. This is included in the door green loop circuit. The status signals are sent to TCMS. The green loop prevents the train from having traction without "all doors closed".
The cab door also has a double contact if door is cut out. This is also supervised by connection to TCMS.
The cab doors are simple passive doors only used by the train staff, they have no door control unit. A double contact can be connected with the "green loop" all doors closed and locked. In the Traction Safe loop is the green loop shown. The cab doors is included in the "green loop" line.
The information on the status of the double contacts has to be sent also to the TCMS.

3.1.15.1.1 Normal condition

The green loop line opens when one door in train opens.

3.1.15.1.2 Failure cases

1) Door is open without opening green loop
2) Door is closed but green loop remains deenergised

3.1.15.1.3 Failure detection

1) TCMS compares signals from door double contacts and the green loop. If they do not have the same status a failure shall be detected, and an event shall be set.
2) All cab doors closed but no traction is possible.

3.1.15.1.4 Failure handling

1) Isolate the door
2) Doors closed and locked bypass to be activated.

Cab door are manually operated.

3.1.15.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.236
Description	If a cab door is open the green loop shall be interrupted
Safety level	SIL 2

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.536	The closed contacts of the cab doors shall be inlcluded in the "all doors closed and locked (green) loop".	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.03.Zefiro-Europe.CONCEPT.1092	Auxiliary contacts used to inform about the status of the cab doors shall be connected to the TCMS by means of digital Input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1289	An event shall be set in case the cab door is not closed	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1093	The status of the cab doors shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

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3.1.16 Driving Mode

3.1.16.1 Requirements

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1547	Entering "Driving Mode" shall be possible only if the following condition are fulfilled: • battery voltage is activated, AND • pantograph is up, AND • main circuit breaker is closed, AND • line voltage is available, AND • 1 cab is active	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_09.03.Zefiro-Europe.CONCEPT.1549	Normal driving mode shall be ended when: • battery voltage is switched off, OR • line voltage is not available, OR • 1 cab is deactivated	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.17 Train Off Mode

3.1.17.1 Requirements

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1548	Train off mode shall be reached from driving mode by the following actions: – by pressing the 'Main circuit breaker off' button – by pressing the 'Pantograph down' button – by deactivating a cab with the key switch – by pressing the "Battery OFF" push button From train off mode the switch-over to driving mode or workshop mode is possible.	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.18 Towing activation

3.1.18.1 Design

Towing mode affects the towed vehicle only, the towing vehicle is not affected by the towing except that the brake control must be done via the UIC pipe by the DBV controller. The braking function in towing mode is defined in the Braking SFRS.

3.1.18.1.1 Normal condition

The train that is about to be towed is coupled mechanically with locomotive or with other train of same type. The cab is then activated in the train to be towed and the Towing switch is then activated. If conditions for "towed" mode is fulfilled the train will be in "towed" mode. In "towed" mode following functions will be blocked in the towed train:

– lifting of pantograph
– closing of LCB
– traction
– dynamic braking

The HVAC and Interior lighting functionality available in "towed" mode will be dependent on the available 3-phase supply through the back-up braking function in the Propulsion system. See HVAC SFSR and Interior lighting SFRS for more information.

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3.1.18.1.2 Failure cases

1) Towing switch is faulty. The train will not be set in "towed" mode. The train must be towed as "dead" train according to operation regulations.
2) "Towed" mode falsely set due to faulty Towing switch. The brake pipe supervision will not open the traction safe loop if the DBV controller is used to brake the train.

3.1.18.1.3 Failure detection

1) An event will be indicated to the driver on the IDU if "towed" mode is activated. The absence of this event will be an indication to the driver that towing is not possible.
2) The Towing switch will be supervised in all inactive cabs and an event will be indicated to the driver if towing switch is activated in an inactive cab.

3.1.18.1.4 Failure handling

1) Towing must be done according to operation regulations. Maintenance required when train is coming to depot.
2) Braking of the train must not be done by the DBV controller. Master controller shall be used instead. This will be managed by the driver according to operation manual. Maintenance required when train is coming to depot.

3.1.18.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.215
Description	The towed V300 Zefiro shall be put into "towed" mode: the cab shall be activated and a dedicated switch for towing shall be activated.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1236	There shall be a five stable position switch on the driver's desk to activate towing: Position 1 - Towing/Pushed Position 2 - Towing/Pulled Position 3 - Towing not active Position 4 - Towed/Pulled Position 5 - Towed/Pushed 1 and 2 are "Towing" positions 4 and 5 are "Towed" positions 	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1237	All the positions of the towing switch shall be notified to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1248	If the Towing switch is in towed position then an event shall be set to indicate that the towing switch is in towed position.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1249	If the Towing switch is in towed position in a de-activated cab then an event shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_09.03.Zefiro-Europe.TRS.217
Description	The switching into "towed" mode shall be interlocked until mechanical coupling is confirmed from the coupler sensor.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.216
Description	The following restrictions apply in towed mode: – brake control via the UIC brake pipe (IP braking), i.e. mechanical brake only. – No traction, line circuit breaker open – no dynamic braking – three phase supply only available from backup braking of the propulsion system (if possible).
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.453	The towed V300 Zefiro shall be put into "towed" mode and an event shall be displayed on the IDU to confirm that "towed" mode is set if - the cab is activated and - the Towing switch is in towed position and - line circuit breaker open and - pantograph down - valid direction is set (train not in neutral) If Towing mode switch is selected in: Treno soccorso (C) Treno soccorso (T)  TCMS shall close load shed contactors (BC2 and BC3)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1580	When towing mode is activated by TCMS a pictogram shall be shown on iDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_09.03.Zefiro-Europe.CONCEPT.457	If Towing switch is turned to towed position and any of following conditions is set then an event shall be set to the driver indicating that towing is not possible: – the cab is not active – line circuit breaker is closed – pantograph is up – direction is not set or not valid	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1238	If "towed" mode is active then all PCU's in the train shall be notified to enable back-up braking (charge DC-link for 3-phase supply).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1239	If "towed" mode is active then the Master controller inputs shall be ignored.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1240	If "towed" mode is active then lifting of any pantograph in the towed train shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1241	If "towed" mode is active then closing of any LCB in the towed train shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1244	If "towed" mode is active then the traction/brake demand to the PCUs in the towed train shall be set to 0.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.526	If towing switch is in towed position and the LCBs are opened and the pantograph is down then the pressure switches on UIC brake pipe in traction safe loop shall be bypassed to allow back up braking of MCM.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL2

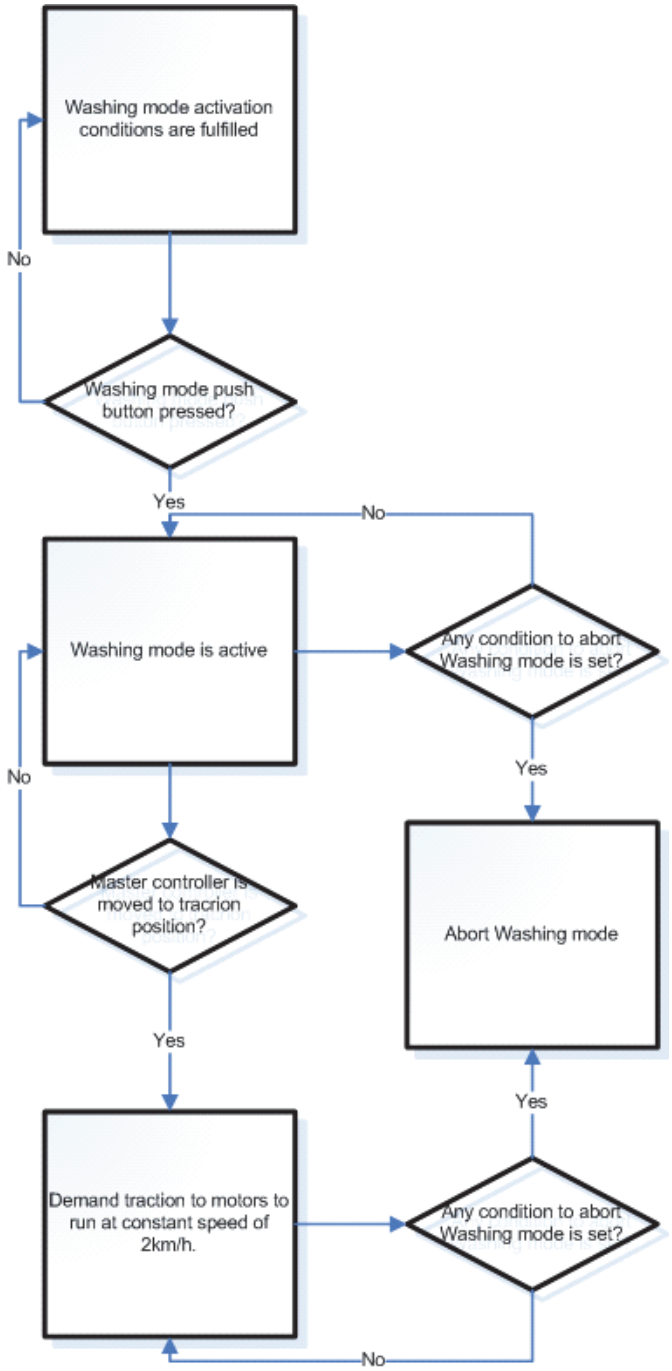
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3.1.19 Washing speed mode

3.1.19.1 Design

Washing mode is used to get a slow steady speed while the train is being washed.
Below flowchart shows how Washing mode is selected/aborted and the washing speed control.



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3.1.19.1.1 Normal condition

If conditions for Washing mode are set then it is possible for the user to activate the Washing mode by pushing a button on the IDU. If Washing mode is successfully activated then the user moves the master controller to any traction position. The train will then start to move at constant speed of 2 km/h (could be changed up to 5 km/h at runtime). The movement of the train and Washing mode will be halted when the master controller is moved to neutral- or brake position.

3.1.19.1.2 Failure cases

1) The speed limitation is not applied even if the washing mode has been selected. Movement through washing plant will be faster than desired.
2) Washing mode is not activated by pressing the dedicated button. Movement through washing plant is not recommended since the train must be driving as normal which could cause the train to move faster than desired through the washing plant.
3) Washing mode is activated without pressing the dedicated button. Train speed will be limited to 2km/h.

3.1.19.1.3 Failure detection

1) Done by the driver/maintainer. Traction block willl be applied and an event will be set.
2) Done by the driver/maintainer via IDU. The washing mode will not be indicated as active on the IDU.
3) Done by the driver/maintainer (Speed reduction noted)

3.1.19.1.4 Failure handling

1) Do not perform the washing procedure, maintenance needed
2) Do not perform the washing procedure, maintenance needed
3) If the train is not in service proceed with maintenance, if the train is in service tow could be needed.

3.1.19.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.218
Description	The Washing mode shall be activated from the IDU.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.285	The IDU shall be equipped with a virtual push button to select the washing mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1140	When the Washing mode is active it shall be indicated on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

ID	2F_09.03.Zefiro-Europe.TRS.82
Description	Washing speed control shall only be possible to selected when: – the Master Controller is in position coast AND – train speed is 0 AND – traction is possible
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.462	Washing mode shall be activated if the washing mode virtual push button is pressed and the following conditions are fulfilled: – the cab is active AND – line voltage is active – the Master Controller is in neutral position (coast) AND – train speed is 0 (standstill) AND – traction is possible (not blocked) – DEP brake available and selected (if not washing speed mode is not allowed)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1104	If washing mode virtual push button is pressed and any of following conditions are true then an event shall be generated: – line voltage is not active – the Master Controller is not in neutral position (coast) – train speed is not standstill – traction is not possible (blocked) – DEP brake is not selected	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.465	The washing mode shall be aborted if any of following conditions are true: – emergency brake active – available traction is less than 25%. – virtual push button is pressed again – brake mode is set to "INDIRECT" (IEP/IP) When master controller (or DBV) is moved in coasting/braking washing mode shall remain active (the train shall go in coasting or shall brake in a normal way)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.397
Description	It shall be possible for the speed at which the train passes through the train washing plant to be adapted to suit each washing plant, i.e. between 2 and 5 km/h.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1137	If washing mode is active and the master controller is moved to any traction position then traction demand shall be calculated to run the train at speed of 2 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1685	It shall be possible to adjust the target speed for washing speed using the speed lever. The allowed values shall be between 2 and 5 km/h. The default value shall be 2 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1103	If the train speed is 2km/h greater than the target speed during washing mode then washing mode shall be aborted.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1138	If the washing speed is aborted due to overspeed then an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1106	If the washing mode is aborted then traction block shall be applied and the traction block shall not be reset until acknowledged with master controller (moved to neutral or braking position).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.20 Coupling speed mode

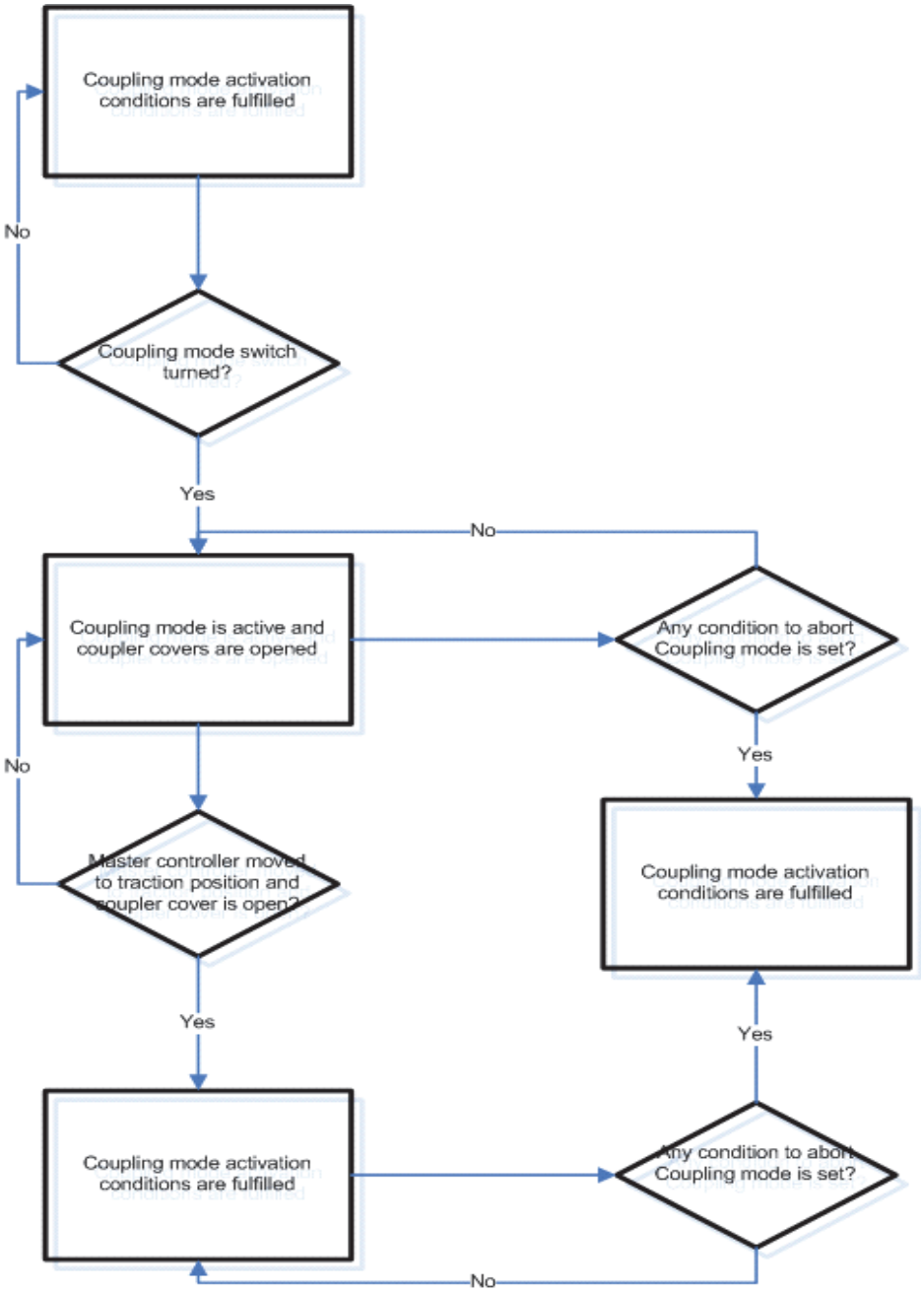
3.1.20.1 Design

In coupling mode one 8 car train is driven towards another 8 car train for multiple coupling. The speed is regulated until mechanical coupling is detected. The speed will be 1 km/h to ensure that the couplers does not take damage.

Below flowchart shows how Coupling mode is selected/aborted and the coupling speed control.

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3.1.20.1.1 Normal condition

If conditions for Coupling mode are set, then it is possible for the user to activate the Coupling mode by pressing the pushbutton on the driver's desk. If Coupling mode is successfully activated, then the coupler covers are opened and then user moves the master controller to any traction position. The train will then start to move at constant speed of 1km/h.
During the coupling sequence, till when coupling is detected, the coupling lamp shall flash
The movement of the train and Coupling mode will be aborted if coupling pushbutton is pressed again or other condition are not fulfilled.
Activation of coupling speed will not be possible when reverse direction is selected.

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3.1.20.1.2 Failure cases

1) The speed limitation is not applied even if the Coupling mode has been selected. The speed will be greater than intended when coupling to the other train.
2) Coupling mode is not activated by turning the Coupling mode switch. The speed must be set manually by the driver.
3) Coupling mode is activated without pressing the dedicated button. Train speed will be limited to 1km/h and coupling covers will be opened.
4) Coupler covers are not opened but indicated as opened. The coupler cover will be damaged.

3.1.20.1.3 Failure detection

1) Done by the driver/maintainer and Traction block will be applied and an event will be set.
2) Done by the driver/maintainer via IDU. The Coupling mode will not be indicated as active on the IDU.
3) Done by the driver/maintainer (Speed reduction noted).
4) Must be ensured by the driver that coupler covers are open. No detection by the train is possible.

3.1.20.1.4 Failure handling

1) Do not perform the coupling procedure, maintenance needed.
2) Do not perform the coupling procedure, maintenance needed.
3) If the train is not in service proceed with maintenance, if the train is in service tow could be needed.
4) Coupling to other train not possible. Multiple operation will not be possible and maintenance is required.

3.1.20.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.219
Description	There shall be a selector switch on the Driver's desk to select coupling speed control.
Safety level	N/A

Nr	**Description**	**Applicability Details**	**Derived to**	**User Interface**	**SIL**
2F_09.03.Zefiro-Europe.CONCEPT.144	There shall be a spring loaded push button on the Driver's desk to select coupling speed control.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1234	When the Coupling mode is active it shall be indicated on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1558	The coupling lamp shall flash when coupling mode is active. When TCMS detects status "train mechanically and electrically coupled" the lamp shall be lit steadly. The lamp shall be switched off after the driver has acknowledged the new train configuration.	Derived Requirement	4S_09.03.05.-.TCMS Software	HW HMI	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1235	If the Coupling speed mode pushbutton is pressed then TCMS shall be notified.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.86
Description	Coupling speed control shall only be possible to be select when: – the Master Controller is in position coast AND – train speed is 0 AND – traction is possible
Safety level	N/A

Front flaps need to be opened manually in the standing train (without active cab).

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Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.295	Coupling mode shall be activated if: – coupling mode pushbtton pressed AND – cab is active AND – the Master Controller is in neutral position (coast) AND – train speed is 0 (standstill) AND – traction is possible (not blocked) AND – line voltage is active AND – Forward direction is selected AND – DEP brake available and selected (if not coupling speed mode is not allowed) AND – Parking Brake not applied	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1220	If coupling mode pushbutton is pressed and any of following conditions are true then an event shall be generated for the driver: – cab is not active – the Master Controller is not in neutral position (coast) – train speed is > 0 km/h – traction is blocked – line voltage is not active – forward direction is not selected – DEP brake is not selected – Parking Brake applied	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.474	Coupling mode shall be aborted if any of following conditions are true: – emergency brake active – coupling is detected (mechanical OR electrical) – push button is pressed again – brake mode is set to "INDIRECT" (IEP/IP) When master controller (or DBV) is moved in coasting/braking coupling mode shall remain active (the train shall go in coasting or shall brake in a normal way)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1683	**GIVEN** (Vehicle is in "*Coupling Mode*") **WHEN** (Both covers of the DM enabled car aren't "*open*" and "*locked*") **THEN** (Traction Block shall be invoked by TCMS)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.670	When coupling mode is active the coupler cover in the front shall be opened.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.323
Description	Minimum speed shall be 0.5 km/h and maximum speed shall be 3km/h.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.641	If the train speed exceeds 3 km/h during coupling mode then coupling mode shall be aborted and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1411	If the coupling mode is aborted then traction block shall be applied and the traction block shall not be reset until until acknowledged with master controller (moved to neutral or braking position).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1193	If coupling mode is active and the couple cover in the front is open and the master controller is moved to any traction position then traction demand shall be calculated to run the train at speed of 1 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.302	If electrical coupling is detected, WTB and ETB inauguration shall start immediately.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

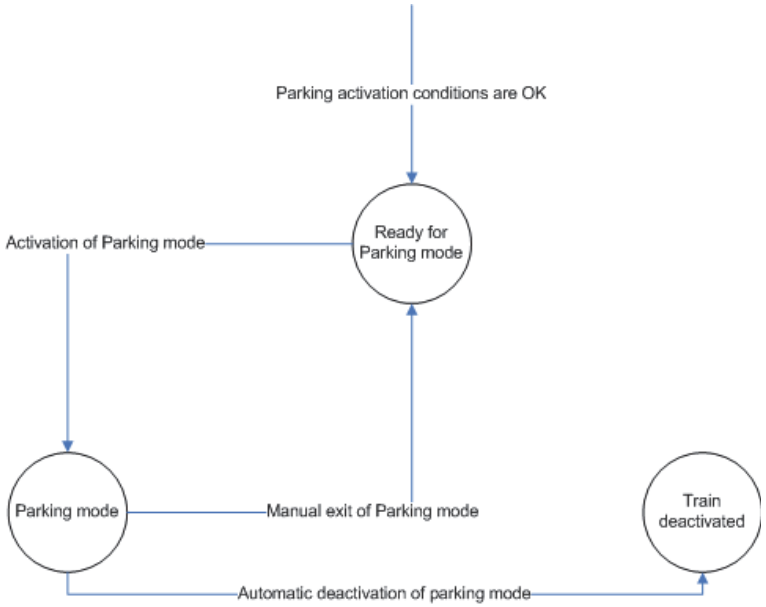
ID	2F_09.03.Zefiro-Europe.TRS.396
Description	A new train configuration shall be always acknowledged by the driver
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1570	Current train configuration shall be displayed on IDU when the train is enabled or everytime the train configuration has changed (e.g copuling/decoupling process, unwanted train separation)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1569	Driver shall acknowledge the train configuration on IDU, otherwise traction block shall be applied by TCMS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.21 Parking Mode

3.1.21.1 Design

Below state diagram defines the fundamentals of the Parking mode



3.1.21.1.1 Normal condition

The Parking mode is activated from the active cab by pressing the Parking push button on driver's desk and at the same time, reset pantographs command and then rotate the driver's key to deactivate the cab. The Parking mode will be confirmed by the illumination of the Parking indicator on driver's desk and an additional indicator visible from outside. All the indicators (internal and external) always show the same state.
Once Parking mode is activated the train will remain all functions that were active before Parking mode. The only difference is that no cab is active and the driver's key is removed.

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The Parking mode will normally be deactivated manually from driver's desk by pressing the Parking push button and at the same time set panto switch according to the wanted configuration and then activate the cab by the driver's key. When Parking mode is manually deactivated the train can be resumed in normal operation.

If any of the pre-conditions for Parking mode activation is no longer present during Parking mode the Parking mode will be deactivated automatically after 10 minutes. The train will in this case be also be deactivated by opening of all battery contactors. There will be a time limit how long Parking mode can be active. When the time limit is exceeded Parking mode and the train will be automatically deactivated.

Deactivation of the parking mode occurs also in case the LCB switch is put in "open" position in one the two cabs. In this case the trainset is immediately deactivated.

The Parking mode shall be possible in single and multiple configuration.

3.1.21.1.2 Failure cases

1) Parking mode is not activated due to faulty Parking pushbutton. The train can not be set in Parking mode which means that line voltage will be deactivated when the driver's key must be removed in order to leave the train.
2) Parking mode can not be manually deactivated due to faulty Parking pushbutton.

3.1.21.1.3 Failure detection

1) Detected by the driver/maintainer by the flashing parking indication lamp.
2) Detected by the driver/maintainer by the steady lit parking indication lamp.

3.1.21.1.4 Failure handling

1) Maintenance required to replace faulty parking pushbutton.
2) The Parking mode can be deactivated by pushing the Emergency stop push button in the cab. This will open all LCB's in the train which in turn will trigger the automatic exit of Parking mode.
   Parking can also be deactivated by activating the cab with drivers key.
   Maintenance is required to replace faulty parking push button at next scheduled maintenance.

3.1.21.2 Requirements

3.1.21.3 Parking mode activation

ID	2F_09.03.Zefiro-Europe.TRS.124
Description	Parking mode shall be entered by: – pressing the 'Parking' push button – keeping the Parking push button pressed and reset panto switches – keeping the Parking push button pressed and rotate and extract the key from the driver's desk – releasing the push button; the rolling stock is in Parking mode and the energy conversion systems must remain active depending on the configuration set on the driver's desk prior to entering into Parking mode.
Safety level	SIL 0

ID	2F_09.03.Zefiro-Europe.TRS.125
Description	The release of the push button before the light on it is lit steady, must interrupt the process for entering into Parking and must configure the trainset based on the new position of the driver's desk levers.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.126
Description	If the configuration is the initial one, the trainset must behave as if the procedure for entering the Parking mode has not been started.
Safety level	N/A

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ID	2F_09.03.Zefiro-Europe.TRS.143
Description	Parking push button shall begin to flash when pressed and shall remain steadily on when the key is removed from the driver's desk.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.140
Description	Conditions for entering the Parking mode: – train in standstill * – Master controller in "0" position on enabled driver desk – Speed controller in "0" position on enabled driver desk – DBV in normal position on enabled driver desk – Train direction set on "netutral" on enabled driver desk – Parking Brake applied * (note: at least 2/3 of parking applied per unit) – main reservoir pressure not below 5.5 bar * – no fire alarm detected * – fire detection system is efficient * – no main circuit breaker trip * – presence of line voltage * – no external power supply inserted * * Conditons above shall be verified on both units in multiple configuration
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.551	Parking mode shall be allowed if: – main reservoir pressure not below 5.5 bar * AND – no fire alarm is active * AND – fire detection system is efficient * AND – no main circuit breaker trip * AND – line voltage is active * AND – train is standstill * AND – Master controller is in coast position on enabled driver desk AND – Speed controller in "0" position on enabled driver desk AND – DBV in normanl position on enabled driver desk AND – Parking Brake applied * (note: at least 2/3 of parking brake applied per unit) – no external power supply inserted * – Train direction set on "netutral" on enabled driver desk * Conditons above shall be verified on both units in multiple configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1263	If the 'Parking' push button is pressed and any of following conditions is set (in single and/or multiple configuration) an advisory event shall be diplayed on the IDU. – main reservoir pressure is below 5.5 bar * – fire alarm is active * – fire detection system is not efficient * – main circuit breaker trip * – line voltage is not active * – train is not standstill * – Master controller is not in coast position on enabled driver desk – Speed controller not in "0" position on enabled driver desk – Parking Brake not applied * (note: at least 2/3 of parking brake applied per unit) – external power supply inserted * – train direction set on "netutral" on enabled driver desk * Conditons above shall be verified on both units in multiple configuration	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.154	Parking mode shall be considered active if all below steps are completed: 1. ‘Parking’ push button is pressed for at least 3 seconds -> The 'Parking' indicators (internal and external) shall start to flash ONLY IF all preconditons are satisfied 2. The 'Parking' push button is still pressed and pantos switches are put in "low" position 3. The 'Parking' push button is still pressed and the driver's key is rotated to deactivate the cab -> The 'Parking' indicators (internal and external) shall be steady lit in all cabs. 4. The 'Parking' push button is released.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.481	The release of the push button before the Parking indicators lit steady, shall interrupt the process for entering into Parking (on both units in case of multiple configuration).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.480	There shall be "parking" push button and lamp in the driver's desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1260	The pressed state of the Parking push button shall be indicated to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1261	When ordered by TCMS the Parking indication (internal and external) lamp shall be illuminated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1501	A set of relays (LCAR) that retain the cab active status during the parking mode shall be provided.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1502	LCAR relays shall be supervised indipendently from the other cab active relays. The series of NC contact and the series of NO contact shall be connected to two different TCMS Inputs.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1503	If a fault/incoherence is detected on LCAR relays an event shall be set and it shall not be possible to activate Parking mode till the event is not solved (by driver or maintenance)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.03.Zefiro-Europe.CONCEPT.1506	It shall be possible to TCMS to activate parking mode circuits in all cab of the trainset.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1507	After the parking mode enter procedure has been confirmed the TCMS shall enable the parking circuits in all the cab of the trainset (also in multiple operation)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

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ID	2F_09.03.Zefiro-Europe.TRS.144
Description	The trainset must have, at the front and back ends, a signal visible from the outside constituted (eg a luminous strip of red LEDs), positioned in the lower centre zone of the front window.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.513	The trainset must have, at the front and back ends, a signal visible from the outside constituted (eg a luminous strip of red LEDs), positioned in the lower centre zone of the front window.	Derived Requirement	4S_08.07.01.-.Drivers Desk	HW HMI	N/A

ID	2F_09.03.Zefiro-Europe.TRS.145
Description	The outside front and back ends visible signal must remain on with a steady light in the Parking mode.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.508	The external 'Parking' indicator shall be steady illuminated when Parking mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1264	When ordered by TCMS the external Parking indication shall be illuminated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.03.Zefiro-Europe.TRS.131
Description	In parking mode the line voltage shall remain activated.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1265	If the cab is deactivated the line voltage shall be deactivated (open all LCB's and lower all pantographs)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1266	If the cab is deactivated and Parking mode is set the line voltage (only in DC mode) shall remain active: – LCB shall remain closed – also the second pantograph (of the same type) shall be raised	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.132
Description	In parking mode, heating, cooling, saloon lighting and passenger door control shall remain in operation.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1267	If the Parking mode is set the external doors shall remain released.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1268	If Parking mode is set the current status of the interior lighting shall remain.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1269	If Parking mode is set the current status of HVAC shall remain.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.528	The PCU shall receive a status signal that parking mode is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.560	To save power and avoid line voltage oscillations: – 50% of LCM/CCM, i.e. all propulsion systems without ACM shall shut down when parking mode is active.	Derived Requirement	4S_05.-.Propulsion		N/A

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3.1.21.4 Manual exit from Parking mode

ID	2F_09.03.Zefiro-Europe.TRS.135
Description	Parking mode shall be finished by: – pressing the 'Parking' push button – keeping the Parking push button pressed and insert and rotate the key to the driver's desk – checking that the light is off and by confirming the exit of the rolling stock from the Parking mode and releasing the push button
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.147
Description	When the operator exits from parking mode by pressing the Parking push button, it shall flash.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.164	Parking mode shall be switched off if following sequnce is completed 1. The 'Parking' push button is pressed for more than 3 seconds (and shall be kept pressed till end of procedure) and pantograph switches are put in "low" position → The 'Parking indicators (internal and external) shall start flashing; 2. The cab is activated by the driver's key → Train configuration confirmation will be required to the driver on IDU; 3. Train configuration is acknowledged on IDU 4. Any Pantograph switch is turned on → TCMS shall acquire as valid panto configuration only if switches status has been stable for 3s; → Once the panto configuration is acquired related LCBs shall be closed (in case of both panto up of the same unit, only rear LCB shall be selected to close); → When LCBs status are confirmed (only one LCB is closed per unit), Panto not selected shall be ordered to lower and 'Parking' indicators (internal and external) shall be switched off in all cabs; 5. The 'Parking' push button is released only if lamp has been switched off This is valid also in case of multiple configuration. Exit parking mode shall be possible from any cab (also from coupled cabs).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1582	If Parking mode push button is released before the end of exit sequence (lamp still flashing), TCMS shall configure the train with all LCB open and according actual driver desk configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		

After manual exit of the Parking mode, in single or multiple configuration, the cab is active and also line voltage is active. Normal operation of the train can be resumed.

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Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1409	The Parking mode shall be aborted if one of the following conditions is present: – The cab is activated without pressing the parking mode pushbutton – More than one key is detected during the enter/exit procedures – Supervision of "parking" relays or "cab active retained" relays detects some faults/incoherences – Train speed greated than 0 km/h – the cab is activated with selected voltage different from catery voltage The TCMS shall re-configure the train opening all LCB and lowering all pantographs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.03.Zefiro-Europe.CONCEPT.1500	If the parking mode is aborted and event shall be set on IDU describing the causes and the not fulfilled conditions	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.21.5 Automatic exit from Parking mode

ID	2F_09.03.Zefiro-Europe.TRS.133
Description	The duration while the train is in parking mode shall be monitored. When the maximum allowed time is reached, the automatic exit shall be triggered (switch to 'train off mode').
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.134
Description	The maximum allowed time shall be a parameter within the TCMS.
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.137
Description	In the absence of at least one of the conditions for entering into Parking mode, the Parking mode must be automatically switched off (on each unit in case of multiple configuration).
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.309
Description	Conditions for leaving the Parking mode: – main reservoir pressure below 5.5 bar * – train speed greater that 0 km/h * – parking brake released * (note: not less than 2/3 of parking brake applied per unit) – fire extinguish system is activated * – fire detection system is not efficient * – main circuit breaker trip * – no line voltage * – external power supply inserted * * Related to each unit in case of multiple configuration
Safety level	N/A

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Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.483	Parking mode shall automatically be switched off if: 1 - main reservoir pressure below 5.5 bar * 2 - main circuit breaker trip (except in case of emergency exit) * 3 - fire extinguish system is active (fire detected) * 4 - fire extinguish system is not efficient * 5 - parking brake released * (note: not less than 2/3 of parking brake applied per unit) 6 - no presence of line voltage * 7 - more than 1 week have elapsed since Parking mode activation * Related to each unit in case of multiple configuration In case condition 3 or 4 is verified, LCB shall be opened before starting the exit timer In case the LCB of a unit of the multiple configuration is undue opened, the LCB of the other unit shall be opened.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.339	The maximum allowed time in Parking mode shall be a parameter stored i NVRAM.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_09.03.Zefiro-Europe.TRS.138
Description	Upon automatic exit from Parking mode, the following must occur: – the start of the 10 minute timed period; – the activation of the internal and external flashing signals;
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.139
Description	During the timed period, the trainset must remain with the panto up and, after 10 minutes, the following must occur, in succession: a) controlled closure (therefore with sensor edge switched on) of all of the access doors, 60 seconds prior to the expiration of the timed period; b) cut-off of the train batteries c) train enters train OFF mode
Safety level	N/A

ID	2F_09.03.Zefiro-Europe.TRS.146
Description	The external Parking signal must flash upon automatic exit from Parking mode (for example when the protections are triggered) and during the timing prior to the cut-off of the train batteries.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.485	If parking mode is pending to be automatically switched off depending on the condtions not beeing fulfilled, this shall be indicated by: – the 'Parking' indicators shall flash.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1272	The automatic shut-down of the train shall be possible to stop if the Parking mode is manually turned off.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.489	When 9 minutes has elapsed since trigger to automatically exit Parking mode was detected the release of the external doors shall be removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_09.03.Zefiro-Europe.CONCEPT.1273	When 10 minutes has elapsed since trigger to automatically exit Parking mode was detected the following shall happen: – all LCB's in the train shall be opened (if not already open) – all pantographs in the train shall be lowered – all battery contactors in the train shall be opened.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.03.Zefiro-Europe.CONCEPT.1571	If Automatic exit procedure is triggered by the following condition: – No Line voltage detected in case during the following 9 minutes the line voltage is detected again and only if all the other preconditions are still satisfied, the automatic procedure shall be stopped and LCB shall be re-closed. Consequently the train status "parking mode" is reactivated. LCB shall be automatically closed after a defined amount of time from the line voltage return that is proportional to the train number (to avoid simultaneous closing of many train LCBs)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.21.6 Emergency Exit

ID	2F_09.03.Zefiro-Europe.TRS.384
Description	If the driver commands the opening of the LCB from one of the cabs, the immediate exit from the parking mode shall be commanded. The train will go into the "disabled" state with all LCB open and all pantographs down
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.513	If the driver commands the opening of the LCB from one of the cabs or any Emergency Push button is pressed from one of the cabs the immediate exit from the parking mode shall be commanded (no timing to disable battery will start) The train will go into the "disabled" state with all LCB open and all pantographs down	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.22 HMI Power Reset

3.1.22.1 Design

If the HMI411 safe processor detects inconsistencies (screen supervision, watch dog or SDT error of input data) it enters the operational safe state (DOSS). From this state it can be recovered by a power cycle. It cannot be recovered from a non operational safe state (NOSS). If CCU-O detects that HMI is in fail safe state an automatic power cycle shall be performed max two times, then the HMI shall be replaced.

3.1.22.1.1 Normal condition

In normal operation the HMI is working properly and TCMS is not commanding any power cycle. The power supply is connected through a relay normally closed contact.
In case an inconsistency is detected TCMS energizes the relays disconnecting HMI power supply

3.1.22.1.2 Failure cases

1) Command relay is kept energized (IDU not supplied)
2) Command relay faulty (power cycle not possible)

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3.1.22.1.3 Failure detection

1) detected by the Driver during operation
2) detected by the Driver during operation

3.1.22.1.4 Failure handling

1) Command relay shall be replaced during maintenance
2) Replace IDU during operation. Command relay to be replaced during maintenance

3.1.22.2 Requirements

ID	2F_09.03.Zefiro-Europe.TRS.395
Description	HMI is in fail safe state shall be automatic recovered by TCMS via Power cycles
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1555	A MIO DO shall control a relay with a normally closed contact in the power supply chain of the IDU (HMI411). The power shall be cut via an active MIO DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		

3.1.23 Train Integration

3.1.23.1 Design

3.1.23.1.1 Normal Condition

3.1.23.1.2 Failure cases

3.1.23.1.3 Failure detection

3.1.23.1.4 Failure handling

3.1.23.2 Requirements

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_09.03.Zefiro-Europe.CONCEPT.1608	When Green Loop is de-energised, TCMS shall command Platform Mode to the all Saloon HVACs	Derived Requirement	4S_09.03.05.-.TCMS Software		

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.28			

4.1.2 TCMS SW

4.1.2.1 Input

Cab Activation/Deactivation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1297	– 'Any cab active' train line – 'Any cab active' train line **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.948 2F_09.03.Zefiro-Europe.CONCEPT.1253
2F_09.03.Zefiro-Europe.CONCEPT.1299	– 'Local cab activated' – 'Local cab activated' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1006
2F_09.03.Zefiro-Europe.CONCEPT.1298	– 'Drivers key in active position' – 'Drivers key in active position' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1006
2F_09.03.Zefiro-Europe.CONCEPT.1300	– 'Drivers key in off position' – 'Drivers key in off position' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1006

Demand of Driving Direction:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1302	– 'Forward selection pushbutton' – 'Forward selection pushbutton' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.969
2F_09.03.Zefiro-Europe.CONCEPT.1303	– 'Neutral selection pushbutton' – 'Neutral selection pushbutton' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.969
2F_09.03.Zefiro-Europe.CONCEPT.1304	– 'Reverse selection pushbutton' – 'Reverse selection pushbutton' **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.969

Traction Brake Reference:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1308	– Encoder 1 bit 1 – Encoder 1 bit 2 – Encoder 1 bit 3 – Encoder 1 bit 4 – Encoder 1 bit 5 – Encoder 1 bit 6	L	2F_09.03.Zefiro-Europe.CONCEPT.1148 2F_09.03.Zefiro-Europe.CONCEPT.1176
2F_09.03.Zefiro-Europe.CONCEPT.1309	– Encoder 2 bit 1 – Encoder 2 bit 2 – Encoder 2 bit 3 – Encoder 2 bit 4 – Encoder 2 bit 5 – Encoder 2 bit 6	L	2F_09.03.Zefiro-Europe.CONCEPT.1148 2F_09.03.Zefiro-Europe.CONCEPT.1176

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2F_09.03.Zefiro-Europe.CONCEPT.1310	– 'Master control EB application	L	2F_09.03.Zefiro-Europe.CONCEPT.1028
2F_09.03.Zefiro-Europe.CONCEPT.1316	– 'Master control back up mode switch	L	2F_09.03.Zefiro-Europe.CONCEPT.1184
2F_09.03.Zefiro-Europe.CONCEPT.1311	– Speed lever not in "+" position – Speed lever not in "+" position **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1167 2F_09.03.Zefiro-Europe.CONCEPT.1395
2F_09.03.Zefiro-Europe.CONCEPT.1312	– Speed lever not in "-" position – Speed lever not in "-" position **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1167 2F_09.03.Zefiro-Europe.CONCEPT.1395
2F_09.03.Zefiro-Europe.CONCEPT.1314	– Speed lever in "0" position – Speed lever in "0" position **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1167 2F_09.03.Zefiro-Europe.CONCEPT.1395
2F_09.03.Zefiro-Europe.CONCEPT.1315	– Speed lever acknowledgment pushbutton – Speed lever acknowledgment pushbutton **(redundant)**	L	2F_09.03.Zefiro-Europe.CONCEPT.1167 2F_09.03.Zefiro-Europe.CONCEPT.1395

Traction Safe:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1404	Brake pipe pressure switch 1		2F_09.03.Zefiro-Europe.CONCEPT.1396
2F_09.03.Zefiro-Europe.CONCEPT.1405	Brake pipe pressure switch 2		2F_09.03.Zefiro-Europe.CONCEPT.1396

Traction Safe Override:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1320	– Door interlocking bypass	L	2F_09.03.Zefiro-Europe.CONCEPT.1117
2F_09.03.Zefiro-Europe.CONCEPT.1321	– Traction safe override	L	2F_09.03.Zefiro-Europe.CONCEPT.1117

Traction Demand in Emergency Mode:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1323	– 'Emergency operation mode' switch	L	2F_09.03.Zefiro-Europe.CONCEPT.1120 2F_09.03.Zefiro-Europe.CONCEPT.1121
2F_09.03.Zefiro-Europe.CONCEPT.1324	– 'Emergency operation mode' trainline	L	2F_09.03.Zefiro-Europe.CONCEPT.1120 2F_09.03.Zefiro-Europe.CONCEPT.1121
2F_09.03.Zefiro-Europe.CONCEPT.1325	– 'Traction' Emergency operation mode trainline	L	2F_09.03.Zefiro-Europe.CONCEPT.1049 2F_09.03.Zefiro-Europe.CONCEPT.1124
2F_09.03.Zefiro-Europe.CONCEPT.1326	– 'Brake' Emergency operation mode trainline	L	2F_09.03.Zefiro-Europe.CONCEPT.1049 2F_09.03.Zefiro-Europe.CONCEPT.1124

Driver Desk supervision:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1329	Lamp Test pushbutton	L	2F_09.03.Zefiro-Europe.CONCEPT.104

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Towing activation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1331	Towing not active selected	L	2F_09.03.Zefiro-Europe.CONCEPT.1236 2F_09.03.Zefiro-Europe.CONCEPT.1237
2F_09.03.Zefiro-Europe.CONCEPT.1332	Towing active selected	L	2F_09.03.Zefiro-Europe.CONCEPT.1236 2F_09.03.Zefiro-Europe.CONCEPT.1237

Coupling speed mode:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1334	Coupling speed switch selected	L	2F_09.03.Zefiro-Europe.CONCEPT.144 2F_09.03.Zefiro-Europe.CONCEPT.1235

Parking mode:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1336	Parking push button selected	L	2F_09.03.Zefiro-Europe.CONCEPT.480 2F_09.03.Zefiro-Europe.CONCEPT.1260
2F_09.03.Zefiro-Europe.CONCEPT.1346	Parking lamp turned on/off	L	2F_09.03.Zefiro-Europe.CONCEPT.480 2F_09.03.Zefiro-Europe.CONCEPT.1261

4.1.2.2 Output:

Serious Fault Indication:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.31	– Serious Fault LED 1 – Serious Fault LED 2 (redundant)	L	2F_09.03.Zefiro-Europe.CONCEPT.1079 2F_09.03.Zefiro-Europe.CONCEPT.1085 2F_09.03.Zefiro-Europe.CONCEPT.1142

4.2 Bus Interface

Propulsion to TCMS SW:
Demanding of driving direction:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.32	Forward trainline input 1		2F_09.03.Zefiro-Europe.CONCEPT.1280
2F_09.03.Zefiro-Europe.CONCEPT.1349	Forward trainline input 2		2F_09.03.Zefiro-Europe.CONCEPT.1280
2F_09.03.Zefiro-Europe.CONCEPT.1350	Reverse trainline input 1		2F_09.03.Zefiro-Europe.CONCEPT.1280
2F_09.03.Zefiro-Europe.CONCEPT.1351	Reverse trainline input 2		2F_09.03.Zefiro-Europe.CONCEPT.1280

Traction Safe:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1352	Traction safe 1 MCM1 is energised		2F_09.03.Zefiro-Europe.CONCEPT.1287

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2F_09.03.Zefiro-Europe.CONCEPT.1353	Traction safe 2 MCM1 is energised		2F_09.03.Zefiro-Europe.CONCEPT.1287
2F_09.03.Zefiro-Europe.CONCEPT.1354	Traction safe 1 MCM2 is energised		2F_09.03.Zefiro-Europe.CONCEPT.1287
2F_09.03.Zefiro-Europe.CONCEPT.1355	Traction safe 2 MCM1 is energised		2F_09.03.Zefiro-Europe.CONCEPT.1287

Train Speed Calculation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1360	Axle speed 1 (axle 1, DCU/M1)		2F_09.03.Zefiro-Europe.CONCEPT.399
2F_09.03.Zefiro-Europe.CONCEPT.1361	Axle speed 2 (axle 1, DCU/M2)		2F_09.03.Zefiro-Europe.CONCEPT.399
2F_09.03.Zefiro-Europe.CONCEPT.1340	DCUM_X speed signal	A	2F_09.03.Zefiro-Europe.CONCEPT.973 2F_09.03.Zefiro-Europe.CONCEPT.1130

TCMS SW to Propulsion:
Traction Brake Reference:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1356	Traction demand		2F_09.03.Zefiro-Europe.CONCEPT.794
2F_09.03.Zefiro-Europe.CONCEPT.1357	Traction is active		2F_09.03.Zefiro-Europe.CONCEPT.795
2F_09.03.Zefiro-Europe.CONCEPT.1358	Braking is active		2F_09.03.Zefiro-Europe.CONCEPT.796

Traction Safe:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1348	Traction safe loop status		2F_09.03.Zefiro-Europe.CONCEPT.1367

Train Speed Calculation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1359	Train speed (for slip/slide control)		2F_09.03.Zefiro-Europe.CONCEPT.860

Speed Limitation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1362	Reference speed for wheel diameter calibration (ATP speed)		2F_09.03.Zefiro-Europe.CONCEPT.1368

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Speed Calibration:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1363	Calibrate wheel diameter		2F_09.03.Zefiro-Europe.CONCEPT.404
2F_09.03.Zefiro-Europe.CONCEPT.1364	Wheel diameter DCU/M1, axle 1		
2F_09.03.Zefiro-Europe.CONCEPT.1365	Wheel diameter DCU/M2, axle 1		

Towing Activation:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.33	Command: enable towing mode		2F_09.03.Zefiro-Europe.CONCEPT.1238

Braking to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_09.03.Zefiro-Europe.CONCEPT.1338	BCU_X speed signal	A	2F_09.03.Zefiro-Europe.CONCEPT.973 2F_09.03.Zefiro-Europe.CONCEPT.1130
2F_09.03.Zefiro-Europe.CONCEPT.1339	WSP_X speed signal	A	2F_09.03.Zefiro-Europe.CONCEPT.973 2F_09.03.Zefiro-Europe.CONCEPT.1130

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5 References

[1] 3EST000225-7909 Drive Control-VF CONCEPT
[2] 3ECP413922-0502 Brake Control-VF CONCEPT

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6 Revision Log

6.1 Revision log 3ECP413922-0903__

Revision:	__		
Date:	**2022-02-09**		
Editor:	**A Memoli**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
			First Issue The document derives from 3ECP413719-0903__ of V300ZEFIRO 14 trains modified with CRs described in the following issues
2F_09.03.ISSUES.-.Zefiro-Europe.177	[2F_09.03.Zefiro-Europe.CONCEPT.1098]: [2F_09.03.Zefiro-Europe.CONCEPT.1685]: [2F_09.03.Zefiro-Europe.CONCEPT.1686]: [2F_09.03.Zefiro-Europe.CONCEPT.1687]: [2F_09.03.Zefiro-Europe.CONCEPT.1688]: [2F_09.03.Zefiro-Europe.TRS.397]:	VZES - Functional: change of the maximum speed value in “washing mode”	CR 213107. The maximum speed value in “washing mode” is 6 km/h. According TSI Loc&Pas, it shall be 5 km/h. It is needed to change the parameter value from 6 to 5 km/h to get homologation.

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